Boundary Selection and Sharp Asymptotics for Exact Burg Mirror Descent

Abstract

We study boundary selection, finite physical length and sharp asymptotics for exact fixed-step Burg mirror descent. For a convex C^2 objective with a unique minimizer, normal Hölder growth and a positive Hessian on the supported-face tangent give a full Euclidean tail $O(k^{-\beta})$, where $\beta > 0$ is the normal-growth exponent. This holds in arbitrary codimension and for every normal KKT-gap pattern. After comparator entry, a convex actual-multiplier estimate gives $F_k = O(k^{-1})$. The complete proof then uses exact weighted dissipation, dyadic normal variation and a supported contraction driven by actual increments. Convex analyticity supplies the growth condition. Independently, every orbit for a neighborhood-analytic objective on the closed triangle converges to a KKT point and has finite length, without convexity. Positive normal powers have finite variation even with degenerate supported curvature; an exact scalar no-overshoot lift yields stationary and moving cancellation classes with exactly two supported coordinates. Selected finer results give normal capture, nonlinear two-clock profiles, tilted endpoints and a finite criterion for active rate types. Globally convex cusp examples and an exact split-clock orbit distinguish these conclusions. All results retain the original ambient step restriction $\eta L < 1$.

Keywords: Burg mirror descent; boundary selection; finite physical length; normal-power variation; scalar cancellation; sharp asymptotics

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1 Problem, mechanisms and scope

We study exact fixed-step Burg mirror descent on the closed standard simplex. Boundary point selection, finite Euclidean length and sharp asymptotics are distinct questions: convergence of objective values alone settles none of the latter two. The common difficulty is that the normal mobility vanishes quadratically, whereas the supported coordinates have a different local dynamics.

Let

$$\Delta = \{x \in \mathbb{R}^n : x_i \geq 0,\ \mathbf{1}^T x = 1\}, \qquad h(x) = -\sum_{i=1}^{n} \log x_i \quad (x \in \operatorname{ri}\Delta).$$

For an admitted ambient Euclidean gradient-Lipschitz bound $L > 0$ on Δ, the exact iteration is

$$x_{k+1} = \operatorname*{arg\,min}_{z \in \operatorname{ri}\Delta} \{\eta \langle \nabla f(x_k), z - x_k \rangle + D_h(z, x_k)\}, \qquad x_0 \in \operatorname{ri}\Delta, \quad 0 < \eta L < 1. \tag{M.1}$$

Here $D_h(z,x) = h(z) - h(x) - \langle \nabla h(x), z - x\rangle$. We retain this ambient step restriction in every chart. Changing an ambient representative by a function vanishing on the simplex hyperplane changes its gradient there by a common-coordinate scalar, absorbed by the multiplier; it does not justify a smaller coordinate-dependent Lipschitz constant in the step restriction. Put $\Delta x_k = x_{k+1} - x_k$ and $\mathcal{L}_k = \sum_{j\geq k} \|\Delta x_j\|_2$.

The paper has four logically distinct blocks.

1. For a convex C^2 objective with a unique minimizer and a positive Hessian on its supported-face tangent, normal Hölder growth implies finite physical length. Section 2 gives the complete proof: exact source–target dissipation, dyadic normal variation and a supported contraction driven by actual normal increments. Its analytic corollary covers every finite codimension and every KKT-gap pattern.

2. On the closed triangle, neighborhood analyticity alone gives all-start point selection, full-simplex KKT convergence and finite length, without convexity or an isolated minimizer. Section 3 states this independent result and its boundary straightness consequence; the analytic division and global cluster arguments are proved in the supplement.

3. Degenerate supported curvature requires another mechanism. Section 4 proves positive-power normal variation and an exact scalar no-overshoot lift. Section 5 supplies stationary and moving cancellation classes with exactly two supported coordinates, including the critical quadratic forcing branch with its separate analytic hypotheses.

4. Sections 6–9 retain finer conclusions not furnished by rectifiability: normal capture and split clocks, nonlinear two-clock profiles, tilted degenerate-supported endpoints and a finite criterion for active rate types.

These blocks keep their own hypotheses. The planar result is nonconvex and two-dimensional; the general higher-dimensional theorem assumes uniqueness and a positive supported Hessian; the structured degenerate lifts have one supported tangent dimension. No general degenerate-face or support-loss theorem is asserted.

1.1 Relation to established convergence mechanisms

The convex Burg comparator gives the established $O(\log k/k)$ value bound [7, Theorem 2, equation (20)]. With a unique minimizer, compactness turns that bound into entry. After entry, an elementary convex estimate using the actual source–target multiplier strengthens the value bound to $O(k^{-1})$ for this fixed Burg-simplex orbit. The geometric proof then controls the sum of literal physical increments through weighted dissipation and the supported lift. Orbit-dependent constants and entry times distinguish this estimate from general-barrier fixed-horizon tightness constructions.

Interior-cluster Bregman length theorems [6, Theorems 17–19] do not directly cover a limit with a forced-zero Burg coordinate, outside the kernel domain. Likewise the ordinary KL relative-error hypothesis [10, Section 2.3 and Theorem 2.9] is not supplied by analyticity alone when the physical normal step has vanishing quadratic mobility. The scaled-KL framework has its own applicability conditions: its Burg half-line example fails the extended scaled-KL property, while its automatic subanalytic result assumes a closed kernel domain [5, Section 3, Theorem 1, Example 2 and Proposition 4; Section 6]. These are precise hypothesis distinctions, not a claim that every Bregman or scaled-KL argument fails for Burg objectives. Our normal estimate uses the exact discrete dissipation budget without dividing by the smallest mobility.

All lengths here are in the original Euclidean coordinates. A coordinate $u_k \to 0$ forces infinite Burg-metric length, since $\sum_{j<k} |\Delta \log u_j| \geq |\log u_k - \log u_0| \to \infty$. Neighborhood analyticity, entire analyticity and semialgebraicity are not interchangeable assumptions. Superseded classifications and other historical profile families are retained in the noncompiled source archive; the present supplement contains the complete dependencies of the active results.

2 Physical length from normal growth

For a nondegenerate supported face, finite physical length follows from a short mechanism: exact weighted dissipation, a dyadic estimate for the literal normal motion, and a supported contraction driven by that actual motion. Normal comparability is unnecessary. Analyticity provides the required growth bound, but the dynamic argument needs only C^2 regularity.

Let x_* be the unique constrained minimizer, $N = \{i : x_{*,i} = 0\}$, $u = x_N$, and $F(x) = f(x) - f(x_*)$. The supported-face tangent is

$$T_A = \{v \in \mathbb{R}^n : \mathbf{1}^T v = 0,\ v_i = 0\ (i \in N)\}.$$

All normal coordinates below are literal simplex coordinates.

Theorem 1 (Physical finite length under normal Hölder growth)**.** *Suppose f is convex and C^2 on an open neighborhood of the closed simplex, has a unique constrained minimizer x_*, and has positive Hessian on T_A. The positive-definiteness condition is vacuous if $T_A = \{0\}$. If $N \neq \varnothing$, suppose that, for some $\beta > 0$ and $C, \varepsilon > 0$,*

$$\|x_N\|_1 \leq C\big(f(x) - f(x_*)\big)^{\beta} \quad \text{for every } x \in \Delta \text{ with } \|x - x_*\| < \varepsilon. \tag{M.2}$$

Then every interior orbit of the exact Burg map (M.1)*, under the original ambient restriction $\eta L < 1$, has finite Euclidean physical length. If $N \neq \varnothing$, its normal and full physical tails satisfy*

$$\sum_{k \geq n} \|u_{k+1} - u_k\|_1 + \mathcal{L}_n \leq C' n^{-\beta} \tag{M.3}$$

for sufficiently large n. The constants may depend on the objective, minimizer, fixed step and initial point. This is an upper bound, not a sharp equivalent or a physical tail-length/distance formula. If N is empty, no growth condition is required and the physical tail decays geometrically after entry.

Corollary 2.1 (Analytic finite length in arbitrary codimension)**.** *Suppose f is convex and real analytic on a two-sided open neighborhood of the closed simplex. If its constrained minimizer is unique and its Hessian is positive on the entire supported-face tangent, every interior exact fixed-step Burg orbit with the original ambient $\eta L < 1$ has finite Euclidean physical length. There is no restriction on the number of missing coordinates or on their normal KKT gaps. Empty supported-tangent and normal blocks are allowed.*

The complete proof follows in this section. The dynamic argument first treats C^2 normal growth; the final static argument supplies that growth for convex analytic objectives, including strict normal gaps.

2.1 Exact weighted dissipation and value entry

This section proves Theorem 1 and Corollary 2.1. Throughout the dynamic argument, f is convex and C^2 on an open neighborhood of the closed standard simplex, its constrained minimizer x_* is unique, and the ambient Euclidean gradient-Lipschitz bound and fixed step satisfy $0 < \eta L < 1$. Put $F = f - f(x_*)$. No normal KKT gap is assumed zero.

For an interior source x and its actual target X, exact optimality and the simplex constraint give

$$\frac{1}{X_i} - \frac{1}{x_i} = \eta(\partial_i f(x) - \mu), \qquad \Delta x_i = -\eta\omega_i(\partial_i f(x) - \mu), \tag{M.4}$$

$$\omega_i = x_i X_i, \qquad \mu = \frac{\sum_i \omega_i \partial_i f(x)}{\sum_i \omega_i}. \tag{M.5}$$

The weights are the actual source–target products in every estimate below. Let $B = \max_\Delta \|\nabla f\|_\infty$ and $R_\eta = 1 + 2\eta B$. The multiplier is a positive weighted average, so $|\partial_i f - \mu| \le 2B$. The identities $x_i/X_i = 1 + \eta x_i(\partial_i f - \mu)$ and $X_i/x_i = 1 - \eta X_i(\partial_i f - \mu)$ give the global ratio bounds

$$R_\eta^{-1} \le X_i/x_i \le R_\eta. \tag{M.6}$$

These bounds hold for every coordinate, including near the boundary.

Lemma 1 (Exact weighted sufficient decrease)**.** *Set*

$$Q(x, X) = \sum_i \frac{(X_i - x_i)^2}{x_i X_i}, \qquad c_\eta = \frac{1}{\eta} - \frac{L}{2} > 0.$$

Then

$$F(x) - F(X) \ge c_\eta Q(x, X). \tag{M.7}$$

In particular, for $N' > n$,

$$\sum_{k=n}^{N'-1} Q(x_k, x_{k+1}) \le \frac{F(x_n) - F(x_{N'})}{c_\eta}. \tag{M.8}$$

Proof. Multiply the first equation in (M.4) by $X_i - x_i$ and sum. Since $\sum_i (X_i - x_i) = 0$, the multiplier cancels and

$$\langle \nabla f(x), X - x \rangle = -\frac{1}{\eta} Q(x, X).$$

The ambient descent lemma on the source–target segment gives

$$F(x) - F(X) \ge \frac{1}{\eta} Q(x, X) - \frac{L}{2} \|X - x\|^2.$$

Every coordinate product $x_i X_i \le 1$, so $Q(x, X) \ge \|X - x\|^2$. This proves (M.7); summation proves (M.8). The stipulated $\eta L < 1$ is retained, even though this one estimate only needs $\eta L < 2$. No eventual coordinate-ratio estimate is used. ∎

Lemma 2 (Global entry to the unique minimizer)**.** *Every interior-start orbit satisfies*

$$F(x_k) \le C_0 \frac{\log k}{k} \quad (k \ge 2), \qquad x_k \longrightarrow x_*. \tag{M.9}$$

Its supported coordinates are uniformly positive on a sufficiently late tail.

Proof. For any interior comparator z, the mirror three-point identity is

$$\eta \langle \nabla f(x), X - z \rangle = D_h(z, x) - D_h(z, X) - D_h(X, x).$$

Convexity at the source and ambient smoothness imply

$$\begin{aligned} \eta\big(f(X) - f(z)\big) &\le D_h(z, x) - D_h(z, X) - D_h(X, x) + \frac{\eta L}{2} \|X - x\|^2 \\ &\le D_h(z, x) - D_h(z, X). \end{aligned}$$

The last step uses $\nabla^2 h = \operatorname{diag}(x_i^{-2}) \succeq I$ on the simplex and $\eta L < 1$. Sum from 0 to $k-1$ and use monotonicity from Lemma 1:

$$\eta k\big(f(x_k) - f(z)\big) \le D_h(z, x_0).$$

Choose the k-dependent interior point $z = (1 - k^{-1})x_* + k^{-1} x_0$. Bounded first derivatives on the compact simplex give $f(z) - f(x_*) = O(k^{-1})$. In the literal Burg formula, each missing coordinate contributes $O(\log k)$ to $D_h(z, x_0)$, while supported terms are bounded. Thus $D_h(z, x_0) = O(\log k)$, proving the value estimate. This is the established convex log-barrier comparator mechanism [7, Theorem 2]. Compactness and uniqueness then imply convergence to x_*, and its positive limiting supported coordinates are protected. ∎

2.1.1 A convex post-entry value estimate

The comparator estimate establishes entry without a length or sharp-rate argument. Convexity now removes its logarithmic loss for this fixed simplex orbit.

Lemma 3 (Inverse-linear objective gap)**.** *Under the standing convex hypotheses,*

$$F(x_k) = O((k+1)^{-1}). \tag{M.10}$$

Neither analyticity nor semialgebraicity nor a supported-Hessian lower bound is needed.

Proof. On a late tail, $x_i \geq x_{*,i}/2$ for every supported coordinate. Set $b_i = (x_i - x_{*,i})/x_i$. For missing coordinates $b_i = 1$, and for supported coordinates $|b_i| \leq 1$. With the actual step multiplier μ, convexity and mass conservation give

$$\begin{aligned} 0 \leq F(x) \leq \langle \nabla f(x), x - x_* \rangle &= \sum_i (x_i - x_{*,i})(\partial_i f(x) - \mu) \\ &\leq \left(\sum_i \frac{(x_i - x_{*,i})^2}{x_i X_i} \right)^{1/2} \left(\sum_i x_i X_i (\partial_i f(x) - \mu)^2 \right)^{1/2} \\ &\leq \sqrt{nR_\eta} \left(\sum_i x_i X_i (\partial_i f(x) - \mu)^2 \right)^{1/2}. \end{aligned}$$

The last line uses $x_i/X_i \leq R_\eta$ and $|b_i| \leq 1$. Since $Q = \eta^2 \sum_i x_i X_i (\partial_i f - \mu)^2$, Lemma 1 yields

$$F_k - F_{k+1} \geq c_* F_k^2, \qquad c_* = \frac{\eta(1 - \eta L/2)}{nR_\eta} > 0.$$

If $F_k = 0$ at a finite step, sufficient decrease forces every later step to vanish. Otherwise $F_k > 0$ and monotonicity gives

$$\frac{1}{F_{k+1}} - \frac{1}{F_k} = \frac{F_k - F_{k+1}}{F_k F_{k+1}} \geq c_*.$$

Summation proves (M.10). The multiplier throughout is the actual source–target multiplier, not a KKT multiplier or an auxiliary weighted mean. ∎

The constants and entry time may depend on the orbit. This fixed Burg-simplex estimate does not contradict the fixed-horizon tightness construction for a general barrier class in [7, Theorem 3]. Uniqueness is used for entry; no claim of automatic support protection for every nonunique problem is made.

2.2 Dyadic control of the literal normal variation

Let $N = \{i : x_{*,i} = 0\}$ and write $u = x_N$. The argument in this section does not use a supported-Hessian hypothesis. Suppose only that $N \neq \varnothing$ and the normal growth bound in Main (M.2) holds for some $\beta > 0$. Write $F_k = F(x_k)$.

Lemma 4 (Normal block and tail bounds)**.** *For all sufficiently late integers n,*

$$\sum_{k=n}^{2n-1} \|u_{k+1} - u_k\|_1 \le C\sqrt{n}\, F_n^{\beta}(F_n - F_{2n})^{1/2} \le Cn^{-\beta}, \tag{M.11}$$

$$\sum_{k\ge n} \|u_{k+1} - u_k\|_1 \le C' n^{-\beta}. \tag{M.12}$$

In particular the literal normal coordinates have finite total variation, without a monotonicity or comparability assumption.

Proof. By Lemma 2, the full tail is in the feasible neighborhood where the growth bound applies. For every $n \le k < 2n$, monotonicity gives both $F_k \le F_n$ and $F_{k+1} \le F_n$. Hence both source and target normal sums are at most CF_n^{β}, and

$$\sum_{k=n}^{2n-1} \sum_{i\in N} u_{i,k} u_{i,k+1} \le CnF_n^{2\beta}.$$

Cauchy–Schwarz over the double index (k, i) and the exact dissipation budget give

$$\begin{aligned}
\sum_{k=n}^{2n-1} \sum_{i\in N} |u_{i,k+1} - u_{i,k}| &\le \left(\sum_{k=n}^{2n-1} \sum_{i\in N} u_{i,k} u_{i,k+1} \right)^{1/2} \\
&\quad \cdot \left(\sum_{k=n}^{2n-1} \sum_{i\in N} \frac{(u_{i,k+1} - u_{i,k})^2}{u_{i,k} u_{i,k+1}} \right)^{1/2} \\
&\le C\sqrt{n}\, F_n^{\beta}(F_n - F_{2n})^{1/2} \\
&\le C\sqrt{n}\, F_n^{1/2+\beta}.
\end{aligned}$$

The normal sum in the second factor is bounded by the full Q. Inserting (M.10) proves (M.11). For $n = 2^j$ its right side is $O(2^{-j\beta})$, which is summable because $\beta > 0$. Finite early segments are harmless.

For the stated tail estimate, partition $[n, \infty)$ into blocks $[2^j n, 2^{j+1} n)$. Their total is at most

$$Cn^{-\beta} \sum_{j\ge 0} 2^{-j\beta} = \frac{C}{1 - 2^{-\beta}} n^{-\beta}.$$

This proves (M.12). ∎

This block estimate uses small literal normal mobility and the total objective decrease together. It never divides by the smallest normal mobility or assumes one common clock. Polynomially divergent ratios, including the exact examples in Proposition S3.1, cause no obstruction.

The proof gives a tail bound, not a pointwise estimate $\|\Delta u_k\| = O(k^{-1-\beta})$. No such pointwise rate is inferred.

2.3 Supported contraction driven by actual normal movement

Assume now that the Hessian is positive on the entire supported-face tangent. We prove that finite normal variation from Lemma 4 lifts to finite physical length. Only C^2 regularity of f is used.

If the support consists of one coordinate s, there is no supported tangent. The simplex constraint gives

$$|\Delta x_s| = \left|\sum_{i\in N} \Delta u_i\right| \le \|\Delta u\|_1,$$

so finite normal variation immediately gives full physical length. Suppose for now that both the supported tangent and the normal block are nonempty.

Choose fixed affine coordinates

$$x = x_* + Bu + Jy, \qquad B_N = I, \quad J_N = 0, \quad \mathbf{1}^T B = 0, \quad \mathbf{1}^T J = 0,$$

where the columns of J span the entire supported tangent. Positive definiteness of $F_{yy}(0,0)$ and the C^1 implicit-function theorem give a C^1 partial minimizing sheet $y = \psi(u)$, with $\psi(0) = 0$. It is locally Lipschitz and gives feasible points for small $u \ge 0$, because the supported coordinates remain positive. Put $e = y - \psi(u)$.

2.3.1 The exact local map and its normal error

Let λ_* be the common supported gradient at x_*. The scalar normalization equation for

$$X_i = \frac{x_i}{1 + \eta x_i(\partial_i f(x) - \mu)} \tag{M.13}$$

has derivative $\eta \sum_{s\notin N} x_{*,s}^2 > 0$ with respect to μ at (x_*, λ_*). Since f is C^2, this equation is C^1. The implicit-function theorem gives a C^1 multiplier branch and local map, agreeing with the unique exact interior Burg step. On a smaller neighborhood all denominators are nonzero and supported coordinates are protected. This argument permits arbitrary zero and positive normal gaps.

Write $U(u,e)$ and $Y(u,e)$ for the target affine coordinates, and set $\delta(u,e) = U(u,e) - u$. Componentwise, the actual normal displacement is

$$\delta_i(u,e) = -\frac{\eta u_i^2 a_i(u,e)}{1 + \eta u_i a_i(u,e)}, \qquad a_i(u,e) = \partial_i f(x(u,\psi(u)+e)) - \mu(x(u,\psi(u)+e)). \tag{M.14}$$

The functions a_i are C^1 with bounded derivatives. The mean-value theorem in e, with the literal factor u_i^2 retained, gives

$$\|\delta(u,e) - \delta(u,0)\|_1 \le CR^2E, \qquad R = \sum_{i\in N} u_i, \tag{M.15}$$

where $E = \|e\|_G$ in the fixed supported norm defined next. This is an estimate for the actual map, not for an arbitrarily perturbed normal recursion.

2.3.2 A fixed supported contraction

Let J_A denote the supported rows of J and set

$$G = J_A^T \operatorname{diag}_{s\notin N}(x_{*,s}^{-2}) J_A, \qquad H = J^T \nabla^2 f(x_*) J.$$

Both matrices are positive definite, and the original ambient bound gives

$$0 \prec H \preceq L J^T J \preceq LG. \tag{M.16}$$

Define the target supported error

$$\mathcal{B}(u,e) = Y(u,e) - \psi(U(u,e)).$$

Its derivative at the origin is

$$D_e\mathcal{B}(0,0) = I - \eta G^{-1} H. \tag{M.17}$$

Indeed the supported linearization of the exact mirror optimality equation is $G(\Delta y) = -\eta H e$ at $u = 0$, while (M.14) makes $D_e U(0,0) = 0$. Thus the differentiated target-sheet term contributes nothing. The matrix in (M.17) is self-adjoint in the G inner product, and (M.16) with $\eta L < 1$ places its eigenvalues strictly between zero and one. Continuity of the derivative, after shrinking only the neighborhood, yields

$$\|\mathcal{B}(u,e) - \mathcal{B}(u,0)\|_G \le \rho E, \qquad \rho < 1. \tag{M.18}$$

No coordinate-dependent step restriction replaces the admitted ambient one.

2.3.3 The same-sign sheet identity

At a sheet source $e = 0$, stationarity along every supported tangent direction gives $\partial_s f(x) = \lambda(u)$ for all supported s. The exact source–target equation is then

$$\Delta x_s = -\eta x_s X_s(\lambda(u) - \mu).$$

All supported increments have the same sign, or all are zero. Therefore their total variation equals the absolute value of their sum. The simplex constraint gives the exact identity and bound

$$\sum_{s\notin N} |\Delta x_s| = \left|\sum_{s\notin N} \Delta x_s\right| = \left|\sum_{i\in N} \Delta u_i\right| \le \|\delta(u,0)\|_1. \tag{M.19}$$

In particular the full physical sheet step is at most $2\|\delta(u,0)\|_1$ in ℓ^1. Fixed affine coordinate bounds and Lipschitz continuity of ψ imply

$$\|\mathcal{B}(u,0)\|_G \le C\|\delta(u,0)\|_1. \tag{M.20}$$

This expresses sheet forcing in terms of its own normal movement, without a reduced-gradient estimate.

Lemma 5 (Actual-increment active lift)**.** *On a sufficiently late tail, there is a fixed $\rho' < 1$ such that*

$$E_{k+1} \le \rho' E_k + C\|u_{k+1} - u_k\|_1. \tag{M.21}$$

If the actual normal increments are summable, then $\sum_k E_k < \infty$ and the physical orbit has finite Euclidean length.

Proof. Combine (M.18), (M.20) and (M.15):

$$\begin{aligned} E_{k+1} &\le \rho E_k + C\|\delta(u_k, 0)\|_1 \\ &\le (\rho + CR_k^2)E_k + C\|u_{k+1} - u_k\|_1. \end{aligned}$$

Since $R_k \to 0$, the coefficient of E_k is eventually bounded by a fixed $\rho' < 1$. This proves (M.21) with the actual off-sheet normal increment as forcing.

For a starting index K in that tail, sum the stable recurrence through an arbitrary later index. Since $E_k \ge 0$, one obtains

$$(1 - \rho') \sum_{k=K+1}^{M} E_k \le \rho' E_K + C \sum_{k=K}^{M} \|u_{k+1} - u_k\|_1.$$

The right side is uniformly bounded when the normal variation is finite, so $\sum E_k < \infty$. Finally, $y_k = \psi(u_k) + e_k$ gives

$$\|y_{k+1} - y_k\| \le C\|u_{k+1} - u_k\|_1 + C(E_k + E_{k+1}).$$

The physical affine chart has bounded norm. Summing proves finite Euclidean physical length. No relative-gradient tracking or summability of an imposed arbitrary error was assumed. ∎

2.3.4 Quantitative physical tails

Set $a_k = \|u_{k+1} - u_k\|_1$ and $A(n) = \sum_{k \ge n} a_k$. Lemma 4 gives $A(n) = O(n^{-\beta})$. We now show why the same exponent passes through the supported recurrence; qualitative summability alone would not give it.

Since $e_k \to 0$, the sequence E_k is bounded. Let $m_n = \lfloor n/2 \rfloor$. Iterating (M.21) from m_n gives

$$E_n \le (\rho')^{n-m_n} E_{m_n} + C \sum_{j=m_n}^{n-1} (\rho')^{n-1-j} a_j \le M(\rho')^{n-m_n} + CA(m_n) = O(n^{-\beta}).$$

Sum the recurrence over a finite interval $n \le k \le T$ and discard the nonnegative terminal term:

$$(1 - \rho') \sum_{k=n}^{T} E_k \le E_n + C \sum_{k=n}^{T} a_k.$$

Letting $T \to \infty$ yields $\sum_{k\geq n} E_k = O(n^{-\beta})$. The Lipschitz sheet and the affine reconstruction give

$$\mathcal{L}_n \leq C\left(A(n) + \sum_{k\geq n} E_k\right) = O(n^{-\beta}). \tag{M.22}$$

For a one-coordinate support the earlier simplex identity gives this bound directly. Thus Theorem 1 holds whenever the normal block is nonempty.

If N is empty, the minimizer is interior and its entire simplex tangent has positive Hessian. The same fixed-metric derivative (M.17) makes the exact map a uniform local contraction. Global entry has already been proved, so distance and the sum of future steps decay geometrically on a tail. A zero-dimensional simplex is trivial. These cases cover all supported-tangent and normal block dimensions.

2.4 Constrained analytic growth and the analytic corollary

The analytic input is a standard compact subanalytic distance inequality, rather than a further dynamic hypothesis.

Lemma 6 (Constrained analytic power growth)**.** *Suppose f is real analytic on an open neighborhood of the closed simplex and has a unique constrained minimizer x_*. There are a finite even integer $m \geq 2$ and constants $c, \varepsilon > 0$ such that*

$$f(x) - f(x_*) \geq c\|x - x_*\|^m \quad (x \in \Delta,\ \|x - x_*\| < \varepsilon). \tag{M.23}$$

The statement does not require a positive supported Hessian or an isolated unconstrained minimum.

Proof. Choose a small closed feasible neighborhood $K = \Delta \cap \overline{B}(x_*, \varepsilon)$ contained in the analytic domain. The continuous functions $F(x) = f(x) - f(x_*)$ and $d(x) = \|x - x_*\|$ have compact subanalytic graphs on K. They are nonnegative there, and uniqueness gives $F^{-1}(0) = d^{-1}(0) = \{x_*\}$. The compact subanalytic Łojasiewicz inequality [13, Theorem 6.4] therefore gives $F(x) \geq c_0 d(x)^q$ on K, for some $q > 0$. Shrink so that $d \leq 1$ and choose an even integer $m \geq \max\{2, q\}$. Then $d^q \geq d^m$, proving (M.23). ∎

Since $x_{*,N} = 0$, equivalence of finite-dimensional norms gives $\|x_N\|_1 \leq CF(x)^{1/m}$. Theorem 1 proves Corollary 2.1, with full physical tail $\mathcal{L}_n = O(n^{-1/m})$ under its positive-supported-Hessian hypothesis. All normal KKT gaps are allowed: the growth inequality is constrained to the simplex and does not require a stationary full-space germ. The exact iteration always uses the original objective.

The same compact semialgebraic inequality supplies a positive growth exponent for a C^2 semialgebraic neighborhood objective with a unique constrained minimizer. By contrast, bare smoothness and uniqueness do not imply power growth. For example, let $b(t) = e^{-1/t^2}$ for $t > 0$ and $b(t) = 0$ for $t \leq 0$, and let $\phi(u) = \int_0^u \int_0^v b(t)\, dt\, dv$. This is a

convex C^∞ function, positive and flat for $u > 0$ approaching zero. On the two-coordinate simplex, $f(x) = \phi(x_2)$ has the unique constrained minimizer $(1, 0)$, but $x_2 \le C\phi(x_2)^\beta$ fails for every $\beta > 0$. This limits the automatic growth assertion, not the inverse-linear objective estimate.

Scope. Normal variation itself does not use a supported-Hessian condition. The positive entire supported Hessian supplies the stable physical lift and its quantitative tail. These and uniqueness are sufficient hypotheses, not claimed necessary conditions for finite length. Nonunique minimizing sets and unrestricted degenerate supported faces remain outside the general theorem. Normal ratios need not be comparable; Proposition S3.1 gives an exact polynomially split-clock orbit with finite length.

3 Planar analytic point selection

The next theorem is independent of the convex unique-minimum result. Its objective is analytic on a neighborhood of the entire closed triangle and need not be convex. The ambient gradient-Lipschitz bound and the original restriction $0 < \eta L < 1$ are retained.

Theorem 2 (Planar analytic selection)**.** *If $n = 3$ and f is real analytic on an open neighborhood of the entire closed simplex, every orbit of* (M.1) *converges to a full-simplex KKT point x_* and has finite physical length. Convexity, isolated critical points, and a prescribed protected tail are not required.*

The full proof is Supplement Theorem S1 and Lemmas S1–S3. Its order is important: global cluster-set exclusion and point selection precede the local analytic half-plane germ argument. Analytic division is needed at edge interiors and endpoints; an interior KL estimate alone does not prove this boundary theorem. If the boundary cluster set were not a singleton, analyticity along an edge gives a factorization $P = u^a H$. A corrected normalized height $H + Ku$ supplies a no-upcrossing argument and forces the whole late orbit into the positive-height regime. A second division $H = A + uJ$, together with exact reciprocal identities, makes the remaining tangential drift telescope and excludes recurrence. Only after this global selection step does the analytic half-plane lemma give local finite length. The critical angular theorem below uses its own convex entry argument and does not depend on this planar chain.

Corollary 3.1 (Planar boundary straightness)**.** *For a nonstationary boundary tail in Theorem 2, the normalized physical steps converge to a unit direction, and $\mathcal{L}_k/\|x_k - x_*\| \to 1$.*

Proof. The local multiplier and exact map extend analytically in a boundary chart. For every fixed linear functional λ of the step, $\lambda(T(x) - x)$ is analytic. Supplement

Lemma S3 gives its eventual sign, including the identically zero alternative. If two different normalized step directions were cluster points, a linear functional strictly separating them would have both signs infinitely often. Therefore the normalized steps converge to a unit vector d_*. Given $\epsilon > 0$, eventually $\langle d_*, \Delta x_j \rangle \geq (1-\epsilon)\|\Delta x_j\|$. Telescoping the already summable tail gives

$$(1-\epsilon)\mathcal{L}_k \leq \langle d_*, x_* - x_k \rangle \leq \|x_* - x_k\| \leq \mathcal{L}_k.$$

Let $k \to \infty$ and then $\epsilon \downarrow 0$. ∎

This consequence is planar and applies only to a nonstationary boundary tail. It does not assert a direction for arbitrary higher-dimensional singular-metric dynamics.

4 Normal powers and an exact scalar lift

Without supported curvature, the normal dissipation still gives more than ordinary bounded variation. Every positive normal power has finite variation. When the support consists of exactly two coordinates, the remaining supported degree of freedom has an ordered frozen-normal map. Its no-overshoot identity makes actual center variation the precise missing input. We prove both ingredients before applying them to cancellation.

4.1 Positive powers of the literal normals

This section retains the standing analytic hypotheses of Main Corollary 2.1: convexity and real analyticity on a two-sided open neighborhood of the closed simplex, a unique constrained minimizer x_*, an interior start, and the original ambient restriction $0 < \eta L < 1$. No supported-Hessian assumption is imposed in this section. Write $F = f - f(x_*)$, let N be the missing coordinates, and put $u = x_N$.

Lemmas 1, 3 and 6, whose proofs do not use positive supported Hessian, give

$$\begin{aligned} &x_k \to x_*, \qquad F_k = O(k^{-1}), \qquad F(x) \geq c\|x - x_*\|^m \quad \text{locally on } \Delta, \\ &F_k - F_{k+1} \geq c_\eta Q_k, \qquad Q_k = \sum_i \frac{(x_{i,k+1} - x_{i,k})^2}{x_{i,k} x_{i,k+1}}, \quad c_\eta = \eta^{-1} - L/2 > 0. \end{aligned} \tag{M.24}$$

Here $m \geq 2$ is finite. Strict gaps are retained in the static growth proof, and every dynamic identity continues to use the original objective and actual multiplier.

Lemma 7 (A global scalar power bound)**.** *For $a, b > 0$ and $\alpha > 0$,*

$$|a^\alpha - b^\alpha| \leq \alpha \max(a,b)^\alpha |\log(a/b)| \leq \alpha \max(a,b)^\alpha \frac{|a-b|}{\sqrt{ab}}. \tag{M.25}$$

Proof. Apply the mean-value bound to $\exp(\alpha t)$ between $\log a$ and $\log b$. With $t = \log(a/b)$, $|a-b|/\sqrt{ab} = 2\sinh(|t|/2) \geq |t|$. No source–target comparability is required. ∎

Proposition 4.1 (Power variation and its normal tail bound)**.** Every positive power of every literal normal coordinate has finite total variation. More precisely, for each fixed $\alpha > 0$ and all sufficiently late n,

$$\sum_{k=n}^{2n-1} \sum_{i \in N} |u_{i,k+1}^{\alpha} - u_{i,k}^{\alpha}| \le C_\alpha \sqrt{n}\, F_n^{\alpha/m} (F_n - F_{2n})^{1/2}, \tag{M.26}$$

$$\sum_{k \ge n} \sum_{i \in N} |u_{i,k+1}^{\alpha} - u_{i,k}^{\alpha}| \le C'_\alpha n^{-\alpha/m}. \tag{M.27}$$

All normal counts and KKT-gap patterns are allowed. If instead only a bound $\|u\|_1 \le CF^\beta$, $\beta > 0$, is available together with the same entry and dissipation, replace α/m by $\alpha\beta$.

Proof. For $n \le k < 2n$, both the source and target normals are bounded by $CF_n^{1/m}$, using monotonicity of F and (M.24). Apply (M.25), then Cauchy–Schwarz over the double index (k, i):

$$\begin{aligned}
\sum_{k=n}^{2n-1} \sum_{i \in N} |\Delta(u_i^{\alpha})_k| &\le C_\alpha F_n^{\alpha/m} \sum_{k=n}^{2n-1} \sum_{i \in N} \frac{|\Delta u_{i,k}|}{\sqrt{u_{i,k} u_{i,k+1}}} \\
&\le C_\alpha F_n^{\alpha/m} \sqrt{|N| n} \left(\sum_{k=n}^{2n-1} Q_k \right)^{1/2} \\
&\le C_\alpha \sqrt{n}\, F_n^{\alpha/m} (F_n - F_{2n})^{1/2}.
\end{aligned}$$

Substitute the value bound to obtain a block estimate $C_\alpha n^{-\alpha/m}$. Sum over $[2^j n, 2^{j+1} n)$ as in Lemma 4; the positive exponent α/m makes the geometric series summable. The finite prefix is harmless. The same calculation proves the stated β version directly from its normal bound. ∎

Lemma 8 (Weighted logarithmic normal budget)**.** *Let $S_k = \sum_{i \in N} \max(u_{i,k}, u_{i,k+1})$. For every $\beta > 0$ and every sufficiently late starting index K,*

$$\sum_{k \ge K} S_k^\beta \sum_{i \in N} |\log(u_{i,k+1}/u_{i,k})| < \infty. \tag{M.28}$$

Its sum on $[n, 2n)$ is at most $C_\beta n^{-\beta/m}$.

Proof. Both endpoints are bounded on the block by $CF_n^{1/m}$. Use $|\log(a/b)| \le |a-b|/\sqrt{ab}$ and the same double-index Cauchy–Schwarz calculation:

$$\sum_{k=n}^{2n-1} S_k^\beta \sum_{i \in N} |\Delta \log u_{i,k}| \le C_\beta F_n^{\beta/m} \sqrt{n}\, (F_n - F_{2n})^{1/2}.$$

Value entry and dyadic summation prove the result. This is a property of the actual Burg orbit; ordinary bounded variation of arbitrary normal inputs would not imply it. ∎

4.2 An exact scalar supported lift

Assume the support of x_* consists of exactly two coordinates, labelled $1, 2$. Write

$$R = \sum_{i \in N} u_i, \qquad x_1 = (1 - R)/2 + y, \qquad x_2 = (1 - R)/2 - y,$$

and $y_* = (x_{*,1} - x_{*,2})/2$. All estimates take place in a protected box about $(0, y_*)$ entered by (M.24). The supported Hessian may vanish.

4.2.1 The unique continuous partial center

The equation $F_y(u, \psi(u)) = 0$ has a unique continuous local solution for every small u. To prove this, choose fixed endpoints $y_* \pm \delta$ in the supported interval. Convexity and the unique face minimum give strictly opposite signs of $F_y(0, y_* \pm \delta)$. Those signs persist for small u, giving a root. If there were two roots, convexity would make $F(u, \cdot)$ constant between them. Analyticity would extend that constancy across the connected local interval, contradicting the endpoint signs. Compactness and uniqueness show continuity and $\psi(u) \to y_*$. The graph, restricted to a small bounded box, is a locally subanalytic analytic-zero graph. No differentiability or Lipschitz property of ψ is asserted in several normal variables.

4.2.2 An ordered frozen-normal map

Define the reciprocal supported coordinate

$$q(u, y) = \frac{1}{x_2} - \frac{1}{x_1}, \qquad q_y = \frac{1}{x_1^2} + \frac{1}{x_2^2} > 0.$$

At fixed u, this maps the supported interval increasingly onto $\mathbb{R}$. Denote its inverse in y by $\mathcal{Q}(u, z)$. This inverse is smooth in the protected box, with bounded normal derivatives. It is unrelated to the dissipation Q_k.

Freeze the normal source and define

$$T_u(y) = \mathcal{Q}\big(u, q(u, y) - \eta F_y(u, y)\big). \tag{M.29}$$

The physical supported direction is $(1, -1, 0, \ldots)$, so on the feasible segment between y and $\psi(u)$,

$$0 \le F_{yy} \le 2L, \qquad q_y \ge 2, \qquad q_y - \eta F_{yy} \ge 2(1 - \eta L) > 0.$$

Thus T_u is increasing and fixes $\psi(u)$. Since F_y has the sign of $y - \psi(u)$, the step moves toward the center, and monotonicity prevents overshoot. Consequently

$$|T_u(y) - \psi(u)| = |y - \psi(u)| - |T_u(y) - y|. \tag{M.30}$$

There is no positive lower bound on F_{yy} or uniform contraction factor.

4.2.3 The simultaneous exact step

Subtract the actual supported reciprocal equations. The multiplier cancels and $F_y = \partial_1 f - \partial_2 f$, giving

$$q(u_{k+1}, y_{k+1}) = q(u_k, y_k) - \eta F_y(u_k, y_k). \tag{M.31}$$

The frozen map and this update have exactly the same dual output. Only inversion at the target normals differs. Bounded derivatives of $\mathcal{Q}$ therefore give

$$|y_{k+1} - T_{u_k}(y_k)| \le C\|u_{k+1} - u_k\|_1. \tag{M.32}$$

This uses actual normal movement, not an imposed gradient perturbation.

Theorem 3 (Scalar lift from actual center variation)**.** *If $\sum_k |\psi(u_{k+1}) - \psi(u_k)| < \infty$, the physical orbit has finite Euclidean length. More precisely, with $e_k = y_k - \psi(u_k)$, a late tail obeys*

$$|\Delta y_k| \le |e_k| - |e_{k+1}| + |\Delta\psi_k| + 2C\|\Delta u_k\|_1. \tag{M.33}$$

For every finite $T \ge K$,

$$Y_{K,T} := \sum_{k=K}^{T} |\Delta y_k| \le A_K + \sum_{k=K}^{T} |\Delta\psi_k|, \quad A_K = |e_K| + 2C \sum_{k\ge K} \|\Delta u_k\|_1 < \infty. \tag{M.34}$$

Proof. Put $d_k = T_{u_k}(y_k) - y_k$ and $\varepsilon_k = y_{k+1} - T_{u_k}(y_k)$. From (M.30),

$$|e_{k+1}| \le |e_k| - |d_k| + |\varepsilon_k| + |\Delta\psi_k|, \qquad |\Delta y_k| \le |d_k| + |\varepsilon_k|.$$

Eliminate $|d_k|$ and apply (M.32) to obtain (M.33). Summing on a finite horizon and dropping the nonpositive terminal error gives (M.34). Ordinary normal variation is finite by Proposition 4.1 with $\alpha = 1$. If center variation is finite, so is supported variation. Finally the affine chart gives

$$\|\Delta x_k\|_2 \le \|\Delta x_k\|_1 \le 2\|\Delta u_k\|_1 + 2|\Delta y_k|,$$

which proves the assertion. ∎

One sufficient condition is that ψ is Lipschitz as a function of $(u_1^{\alpha_1}, \dots, u_q^{\alpha_q})$ for positive exponents. Proposition 4.1 supplies the needed variation. Another sufficient condition, where the derivatives exist in the positive box, is

$$|u_i \partial_i \psi(u)| \le C R^{\alpha} \quad \text{for every } i, \qquad \alpha > 0.$$

Interpolate each consecutive normal coordinate geometrically. Its logarithmic derivative is $\log(U_i/u_i)$, while the interpolated mass is at most $\sum_i \max(u_i, U_i)$. Integrating the derivative estimate and applying Lemma 8 proves finite center variation. This condition is not automatic at cancellation cusps. Neither ordinary normal variation nor arbitrary Hölder continuity of a multivariate center is sufficient by itself.

4.2.4 One normal as an alternative proof

With a single normal u, the continuous bounded subanalytic center has a convergent one-variable Puiseux expansion on a right-hand interval [11, Proposition 1.8.4]. After subtracting its limit, either it is constant or its first exponent α is positive. Termwise differentiation yields

$$|\psi'(u)| \le Cu^{\alpha-1}, \qquad |\psi(v) - \psi(u)| \le C'|v^\alpha - u^\alpha|.$$

Proposition 4.1 and Theorem 3 prove length without positive supported curvature or monotone normals. This is an alternative proof of an already covered one-normal case, not additional class coverage. No multiparameter Puiseux assertion is used.

5 Structured degenerate-supported cancellation

5.1 Stationary pair and cluster contrasts

For the remainder of this section the support of x_* consists of *exactly two* coordinates, labelled $1, 2$. With normal vector $n = x_N$ and $R = \sum_{i\in N} n_i$, write

$$x_1 = (1 - R)/2 + y, \qquad x_2 = (1 - R)/2 - y.$$

The scalar partial center $\psi(n)$ is the unique continuous local solution of $F_y(n, \psi(n)) = 0$. Convexity, analyticity and the unique face minimum guarantee its existence and uniqueness; several normal parameters need not make it Lipschitz.

Theorem 4 (Scalar center variation and stationary cancellation)**.** *Retain the convex analytic, unique-minimum and original ambient-step hypotheses of Section 4. The orbit has finite physical length for either of the following local objective forms, with any finite number of extra literal normals:*

1. *A stationary pair contrast. For two selected normals u, v and the remaining normals ξ,*

$$F = C(u - v, y) + B(u + v, \xi) - c_*, \qquad C_r(0, y) = 0. \tag{M.35}$$

   Here C, B are analytic and the total objective has the standing convexity and unique-minimum properties.

2. *A stationary cluster contrast. For selected normals $v_1, \dots, v_h$, $h \ge 2$, put*

$$S = \sum_{i=1}^{h} v_i, \qquad W = \sum_{i<j} (v_i - v_j)^2.$$

   The objective has the form

$$F = \Phi(W, y) + G(S, \xi) - c_* \tag{M.36}$$

   with analytic Φ, G and the same total-objective hypotheses.

In the pair class every $|u-v|^\alpha$, $\alpha>0$, has finite variation; in the cluster class this holds for every $|v_i-v_j|^\alpha$ and W^α. Common selected-normal strict gaps and arbitrary extra-normal gaps are allowed when the stated objective form holds. Separate convexity of the summands is sufficient but is not required.

Theorem 3 reduces the conclusion to actual center variation. The proof is a finite-horizon telescope, not a uniform-contraction argument.

For the pair class, analyticity and convexity give $C_r = rb$, $b \geq 0$. The exact reciprocal contrast $z = 1/u - 1/v$ satisfies

$$z^+ = z(1 - 2\eta uvb).$$

Its late multiplier lies in $(0, 1]$. Every $|z|^\alpha$ is then nonincreasing, and $|u-v|^\alpha = |z|^\alpha u^\alpha v^\alpha$ has finite variation by Proposition 4.1. The scalar center depends only on $u - v$; a one-variable Puiseux expansion on its eventual sign controls its variation by one positive contrast power. For the cluster, the same argument uses

$$(1/v_i - 1/v_j)^+ = (1/v_i - 1/v_j)(1 - 2h\eta v_i v_j \Phi_W), \qquad \Phi_W \geq 0.$$

Every W^α has finite variation, and the center depends only on W. Supplement Sections S2.1–S2.2 prove the cancellation identities and the center-variation conclusions. The one-normal Puiseux specialization is an alternative proof of an already covered case. Arbitrary Hölder continuity of a multivariate center would not suffice for the lift.

5.2 A moving one-signed contrast

The next class does not preserve a stationary normal cancellation sheet. Let $n = (u, v, \xi)$ be the literal normal vector, $r = u - v$ and $s = u + v$.

Theorem 5 (Moving contrast with positive transverse restoration)**.** *Retain the standing assumptions and exactly two supported coordinates. Suppose locally*

$$F = C(r, y) + B(n) - c_*, \qquad C_r(r, y) = rb(r, y), \qquad 0 < b_0 \leq b(r, y) \leq b_1, \tag{M.37}$$

where the functions are analytic on protected neighborhoods. After exchanging u, v if needed, assume that for some $\nu > 2$ the force $h = B_v - B_u$ satisfies, uniformly throughout a sufficiently small positive normal box,

$$cs^\nu \leq h(n) \leq Cs^\nu, \qquad |n_j \partial_j h(n)| \leq Cs^\nu \quad \text{for every literal normal } n_j. \tag{M.38}$$

Then the orbit has finite Euclidean physical length. The reciprocal contrast $z_k = 1/u_k - 1/v_k$ is bounded on a tail, has at most one sign crossing there, and $u_k/v_k \to 1$. Every $|z_k|^\alpha$ and $|u_k - v_k|^\alpha$, $\alpha > 0$, has finite total variation. The extra-normal count is arbitrary. No monotonicity of individual normals, or finite-crossing property for the moving normal stationary sheet, is asserted.

Here is the mechanism, with the order of the estimates important. The exact recurrence is $z^+ = (1-a)z - g$, where $a = 2\eta uvb$ and $g = \eta h > 0$. Once $z < 0$, set $t = -z$; then

$$t^+ = (1-a)t + aq, \qquad q = \frac{h}{2uvb}.$$

Before using comparability, the identity $r = uvt$ gives

$$\frac{s^\nu}{uv} = s^{\nu-2}(4 + rt) \le 4s^{\nu-2} + s^{\nu-1}t.$$

This gives a bounded reciprocal barrier and only then $u/v \to 1$. For a positive power α, logarithmic differentiation of q^α and (M.28) control its normal variation. Its remaining supported-motion term is at most $C\delta^{\alpha(\nu-2)} \sum |\Delta y|$ on a tail with $s \le \delta$. Since t^+ lies between t and q, an ordered-power telescope and the exact product $r^\alpha = (uv)^\alpha t^\alpha$ give, on every finite horizon,

$$\sum |\Delta(r^\alpha)| \le K + C\delta^{\alpha\nu} \sum |\Delta y|,$$

where K is already known finite. Choose the one Puiseux power controlling the scalar center, combine this bound with (M.33), and absorb the last term for sufficiently small δ. The horizon tends to infinity only after absorption. All positive reciprocal powers follow afterwards. Supplement Section S2.3 supplies the full proof, including the other sign branch. The hypotheses $\nu > 2$ and $b \ge b_0 > 0$ are retained; sign-changing or vanishing-restoration forcing is not covered.

5.3 A globally convex cusp family

These hypotheses permit a genuinely degenerate supported minimum and a non-Lipschitz center. Set $\varepsilon = 1/100$ and

$$C_0(r, y) = r^2(1 + 2\varepsilon y + 5\varepsilon^2 y^2) + y^4 + \frac{25}{3}\varepsilon^4 r^4. \tag{M.39}$$

This polynomial is globally convex. For every integer $M \ge 2$, both ambient polynomials

$$C_0(x_3 - x_4, (x_1 - x_2)/2) + (x_3 + x_4)^{2M}, \qquad C_0(x_3 - x_4, (x_1 - x_2)/2) + (x_3 + 2x_4)^{2M}$$

have the unique simplex minimizer $(1/2, 1/2, 0, 0)$, zero supported Hessian and zero normal gaps. Their supported center satisfies $\psi(r) \sim -(\varepsilon/2)^{1/3}|r|^{2/3}$. The first fits Theorem 4; the second fits Theorem 5 with $\nu = 2M - 1$ and has a genuinely moving normal stationary sheet. Thus all these degrees have finite physical length. Supplement Section S2.4 gives the exact global Hessian certificate, the cusp proof, arbitrary-cluster extensions and the moving-sheet expansion.

The normal-power proposition is unrestricted in supported dimension, whereas the new degenerate-supported lifts and cancellation classes have one supported tangent dimension. They do not establish a general degenerate-face theorem or a nonunique-minimizer theorem. The coupled-jet and mixed-rate results below retain their different hypotheses and stronger asymptotic conclusions.

5.4 The critical quadratic forcing endpoint

The positive weighted-logarithmic exponent in Theorem 5 vanishes at forcing order two. Under the full analytic neighborhood hypotheses, a different bounded-contrast chart closes that endpoint. The two forcing orders therefore use distinct estimates.

Theorem 6 (Quadratic one-signed forcing)**.** *Let f be convex and real analytic on a two-sided open neighborhood of the closed simplex, with a unique constrained minimizer supported on exactly two coordinates. Write the literal normal vector as $n = (u, v, \xi)$, where the finite number of extra normals is arbitrary, and put*

$$r = u - v, \quad s = u + v, \quad R = \sum_j n_j, \quad x_1 = (1 - R)/2 + y, \quad x_2 = (1 - R)/2 - y.$$

Suppose locally the normalized objective has the analytic separated form

$$F = C(r, y) + B(u, v, \xi) - c_*, \qquad C_r = rb(r, y), \qquad 0 < b_0 \le b \le b_1. \tag{M.40}$$

After exchanging u, v if necessary, assume the uniform independent-box bound

$$c(u + v)^2 \le B_v(u, v, \xi) - B_u(u, v, \xi) \le C(u + v)^2 \tag{M.41}$$

for every sufficiently small positive u, v and independently varying positive extra normals ξ. Then every interior exact Burg orbit with the original ambient $\eta L < 1$ has finite Euclidean physical length. Its reciprocal contrast $z_k = 1/u_k - 1/v_k$ is bounded on a tail, has at most one sign crossing there, and $u_k/v_k \to 1$. Every positive power of $|z_k|$ and of $|u_k - v_k|$ has finite total variation.

No supported-Hessian assumption is imposed. Total convexity suffices; separate convexity of C, B is not required. The selected pair may have equal zero or equal strict KKT gaps; extra-normal gaps are unrestricted subject to the stated structure. No orbit rate, prescribed reciprocal limit or finite-crossing assertion for a moving stationary sheet is claimed.

Why the endpoint closes. Let $h = B_v - B_u$ and $a = 2\eta uvb$. The exact recurrence remains $z^+ = (1 - a)z - \eta h$. The forever-nonnegative branch is handled by monotone reciprocal powers. On the other branch $t = -z > 0$ satisfies

$$t^+ = (1 - a)t + aq, \qquad q = \frac{h}{2uvb}.$$

Before any comparability, $s^2/(uv) = 4 + rt \le 4 + st$ gives a finite upper barrier for t. The lower bound $q \ge 2c/b_1$ then fixes an orbit-dependent interval $0 < m_0 \le t \le M_0$ before the tail is made smaller.

Uniform analyticity in the independent normal box forces the degree-zero and degree-one Taylor coefficients of h in u, v to vanish identically as functions of ξ. In the exact chart

$$d = 2 + \sqrt{4 + s^2 t^2}, \qquad r = ts^2/d, \qquad uv = s^2/d,$$

one has $h = s^2 H(s, t, \xi)$ with analytic H and $H(0, t, \xi)$ independent of t. Thus $q = dH/(2b)$ is analytic through $s = 0$, with bounded derivatives and $q_t = O(s)$ on the fixed protected box.

For each finite horizon put $X = \sum |\Delta t|$ and $Y = \sum |\Delta y|$. The ordered scalar recurrence and normal variation give

$$X \le A + C\delta X + BY \quad (s \le \delta).$$

After the first absorption, $X \le A' + B'Y$. The scalar center is controlled by one Puiseux power r^α, while $r^\alpha = s^{2\alpha}(t/d)^\alpha$ has a smooth bounded second factor because $t \ge m_0$. Every positive power of s has finite variation. The exact scalar supported lift therefore gives

$$Y \le E + D\delta^{2\alpha} X.$$

All displayed additive constants are already finite and independent of the horizon. Keeping m_0, M_0 and the derivative constants fixed, a second tail shrinkage absorbs the resulting Y term. Only afterwards is the horizon sent to infinity. Supplement Section S2.6, p. 21, proves the full analytic divisibility and both absorptions. This argument does not substitute zero into the weighted-logarithmic exponent of the earlier proof.

A global polynomial with genuinely quadratic force. For fixed $e, d > 0$ define

$$C_e(r, y) = r^2(1 + 2ey + 5e^2y^2) + y^4 + (25e^4/3)r^4,$$
$$D_d(s, r) = s^2(1 - 2dr + 5d^2r^2) + r^4 + (25d^4/3)s^4.$$

The ambient polynomial

$$f(x) = C_e(x_3 - x_4, (x_1 - x_2)/2) + D_d(x_3 + x_4, x_3 - x_4) \tag{M.42}$$

is globally convex and has the unique simplex minimizer $(1/2, 1/2, 0, 0)$, zero normal gaps and a vanishing supported Hessian. Its exact force is

$$h = 4ds^2 - 20d^2rs^2 - 8r^3,$$

which is between $2ds^2$ and $6ds^2$ on a sufficiently small positive normal box. Its supported center retains the cusp $\psi(r) \sim -(e/2)^{1/3}|r|^{2/3}$, and its normal stationary sheet moves with leading term $2ds^2/b(0, y)$. Supplement Section S2.7, p. 26, proves global convexity, every parameter case, the exact force and the stationary-sheet expansion. This family falls outside the earlier $\nu > 2$ force bound, without implying any orbit asymptotic formula.

The positive-supported-Hessian theorem remains the general arbitrary-dimensional result. The two moving-contrast theorems are structured scalar alternatives: exactly two supported coordinates, positive restoration, a separated center and one-signed uniform forcing remain explicit. Bounds only along an unknown orbit, arbitrary multivariate centers and general degenerate supported blocks are not covered.

6 Additional geometry from analytic flag capture

Corollary 2.1 gives finite length without normal balance. Convex analytic flags provide a stronger capture conclusion in an oriented zero-gap class. The criterion permits a leading zero plane and nested transverse orders without prescribing a limiting ray.

Let x_* be the unique minimizer, let N be its $q \geq 1$ missing coordinates, and put

$$V = \{v \in \mathbb{R}^n : \mathbf{1}^T v = 0\}, \qquad C = \{v \in V : v_i \geq 0 \ (i \in N)\}, \qquad F(v) = f(x_* + v) - f(x_*).$$

For this section all normal KKT gaps are zero, so $\nabla_V F(0) = 0$ and F is nonnegative on a full local convex neighborhood. Starting with $V_0 = V$, let P_j be the first nonzero homogeneous term of $F|_{V_j}$, of degree m_j, and set

$$V_{j+1} = \{v \in V_j : P_j(v) = 0\}. \tag{M.43}$$

Stop when the restriction is identically zero. Convexity makes these proper nested linear spaces and the degrees strictly increasing and even. The terminal space W satisfies $W \cap C = \{0\}$ by uniqueness. Thus the last cone-visible stage j_* is defined by

$$V_{j_*} \cap C \neq \{0\}, \qquad V_{j_*+1} \cap C = \{0\}. \tag{M.44}$$

The construction and its finite termination, including an analytically flat W, are proved in Supplement Section S3.2.

Theorem 7 (Uniform normal capture from analytic flags)**.** *Suppose f is convex and real analytic on a two-sided open neighborhood of the closed simplex. Suppose its unique constrained minimizer has $q \geq 1$ zero coordinates and positive Hessian on the entire supported-face tangent; an empty supported tangent is allowed. Assume all q normal KKT gaps are zero. If the space V_{j_*} in* (M.44) *contains a fixed physical tangent vector a with*

$$a_i > 0 \qquad (i \in N),$$

then, with $m = m_{j_*}$, every interior orbit of the exact map (M.1) *under the original ambient restriction $\eta L < 1$ has literal normal coordinates satisfying, for all sufficiently large k,*

$$c k^{-1/m} \leq u_{i,k} \leq C k^{-1/m}, \quad i \in N, \tag{M.45}$$

$$\kappa_k := \frac{\max_{i \in N} u_{i,k}}{\min_{i \in N} u_{i,k}} \leq C. \tag{M.46}$$

There is no restriction on the fixed finite value of q. The constants may depend on q and f. These are comparability bounds, not a limiting ray, sharp equivalent or tail-length/distance formula. In addition, $F(x_k) = \Theta(k^{-1})$ and $\mathcal{L}_k = O(k^{-1/m})$.

Capture mechanism. The post-entry convex estimate gives $F(v_k) = O(k^{-1})$. The static flag yields $F(v) \geq c\|v\|^m$ on the feasible cone and $|D_a F(v)| \leq C F(v)^{1-1/m}$ on a full neighborhood. Pairing the exact reciprocal update with the fixed physical vector $a \in V$ cancels the multiplier:

$$Q(x) = \sum_i \frac{a_i}{x_i}, \qquad Q(x_{k+1}) - Q(x_k) = \eta D_a F(v_k).$$

The supported terms are bounded, while every normal reciprocal coefficient is positive. Summation gives the lower bounds in (M.45); cone coercivity gives the upper bounds. Their quotient is bounded. Cone coercivity and the lower normal bound also give $F(v_k) \geq c/k$; Theorem 1 with $\beta = 1/m$ gives the physical tail bound. Supplement Sections S3.1–S3.3 provide the full proof, including flat terminal directions and uniform weighted estimates. Qualitative length already follows without the flag orientation from Corollary 2.1. ∎

6.1 Three-normal examples and a split-clock obstruction

At a vertex of the four-coordinate simplex write $u = (x_1, x_2, x_3)$, with supported coordinate x_0. The globally convex polynomial

$$f(u) = (u_1 + u_2 - u_3)^4 + u_1^6 + u_2^6 + u_3^6$$

has a leading zero plane containing the positive normal vector $(1, 1, 2)$; its next restricted term is positive definite. Theorem 7 therefore gives uniform normal comparability without selecting a ray in that plane. For

$$f(u) = (u_1 - u_2)^2 + (u_1 + u_2 - 2u_3)^4 + (u_1 + u_2 + u_3)^6,$$

the successive orders are $2, 4, 6$, and the last visible space is the positive line $u_1 = u_2 = u_3$. The theorem retains these unequal transverse orders. A positive quadratic supported block may be appended in either example.

The flag condition cannot be omitted from the proposed uniform capture conclusion. For

$$f(u) = (u_1 - u_2)^2 + (u_1 + u_2)^6 + u_3^4, \tag{M.47}$$

the exact symmetry-preserved orbit $u_1 = u_2 = a$, $u_3 = b$ satisfies

$$a_k \sim (1152\eta k)^{-1/6}, \qquad b_k \sim (16\eta k)^{-1/4}, \qquad \frac{a_k}{b_k} \asymp k^{1/12}.$$

Supplement Section S3.5 derives these equivalents directly from the exact multiplier and corrected reciprocal equations. The last visible line lies in $u_3 = 0$, so it contains no strictly normal-positive vector. This orbit disproves every universal fixed-power polylogarithmic balance bound. It has finite physical length and is not a counterexample to rectifiability.

6.2 Capture and the broader length conclusion

The flag orientation is a sufficient condition for uniform capture. It is not required for qualitative analytic finite length: Corollary 2.1 covers coordinate-face last-visible flags, all normal counts and every strict-gap pattern under uniqueness and a positive entire supported Hessian. The earlier two-normal theorem keeps its stronger conditional rates. Degenerate supported Hessians, nonunique minimizing sets and objectives without an appropriate growth assumption remain outside the general theorem.

7 Nonlinear two-clock cancellation

The general theorem gives rectifiability; the result here derives a normal ray, an explicit radial equivalent and a summable step rate by tracking a nonlinear center on two distinct clocks.

Let x_* be the unique constrained minimizer, with support A and exactly two missing coordinates $N = \{i_1, i_2\}$. Write $u = (x_{i_1}, x_{i_2})$ for the literal normal coordinates. In a fixed affine chart

$$x = x_* + Bu + Jy, \qquad B_N = I_2, \quad J_N = 0, \quad \mathbf{1}^T B = 0, \quad \mathbf{1}^T J = 0,$$

the columns of J span the entire supported-face tangent. Assume its Hessian $H = J^T \nabla^2 f(x_*) J$ is positive definite. If this tangent is empty, all active variables and positive-definiteness conditions are omitted. The analytic active minimizer $y = \psi(u)$ defines

$$\phi(u) = f(x_* + Bu + J\psi(u)) - f(x_*), \qquad \gamma = \nabla\phi(0), \quad S = \nabla^2\phi(0).$$

Here $\gamma \geq 0$ consists of the two normal KKT gaps and $S \succeq 0$ is the reduced normal Hessian.

Theorem 8 (Quantitative analytic cancellation)**.** *Suppose f is convex and real analytic on a two-sided open neighborhood of the closed simplex. Assume that its constrained minimizer x_* is unique, has exactly two zero coordinates, and has positive Hessian on its entire supported-face tangent. Assume in addition*

$$\gamma = 0, \qquad \operatorname{rank} S = 1, \qquad \ker S = \operatorname{span}\{a\}, \quad a = (a_1, a_2) > 0.$$

Every interior exact Burg orbit under the original ambient $\eta L < 1$ has the following asymptotics. Set $\ell_i = 1/a_i$, $r = \ell^T u/2$ and $w = (I - a\ell^T/2)u$. Complete the active quadratic square by a fixed affine shear, and minimize all stable variables (w, e) at fixed r. Their analytic center $\Phi(r)$ has a nonconstant reduced value

$$f_0(r) = cr^m + O(r^{m+1}), \qquad c > 0, \quad m \geq 4 \text{ even.} \tag{M.48}$$

Under these additional hypotheses the order m is automatic, and along every such orbit,

$$\frac{u_{i,k}}{r_k} \longrightarrow a_i, \qquad r_k \sim \left(\frac{2}{\eta m^2 c k}\right)^{1/m}, \qquad \|x_{k+1} - x_k\| = O(k^{-1-1/m}). \tag{M.49}$$

These estimates assert neither a sharp physical-step vector nor a tail-length/distance equivalent.

Mechanism and proof location. The theorem has its complete two-clock proof in Supplement Sections S4.1–S4.5, including vector-valued supported tangents. The value bound $f(x_k) - f(x_*) = O(\log k/k)$ and uniqueness first give entry and supported-coordinate protection. Two-sided convexity and analyticity imply the automatic center estimate

$$\Phi'(r) = O(r^{m/2}), \qquad \Phi(r) = O(r^{m/2+1}).$$

Pairing the exact reciprocal update with the fixed physical Hessian-null direction gives separate lower bounds for both literal normals. This proves $u_i/r \to a_i$ before any normal ratio is used in tracking.

Write $W(r)$ for the normal component of $\Phi(r)$, $\delta = w - W(r)$, and $\epsilon = e - \psi(r, w)$ for error from the active center at the current normal point. In the supported Burg norm G and fixed D_a^{-2} normal norm $*$, the exact source–target map then gives

$$\begin{aligned} \|\epsilon^+\|_G &\le \rho\|\epsilon\|_G + Cr^2\|\delta\| + Cr^{m+1}, \\ \|\delta^+\|_* &\le (1 - c_0 r^2)\|\delta\|_* + Cr^2\|\epsilon\|_G + Cr^{m+1}, \qquad \rho < 1. \end{aligned}$$

Together with $|\Delta r|/r = o(r^2)$, these inequalities yield $\delta = O(r^{m-1})$ and $\epsilon = O(r^{m+1})$. The exact reciprocal identity proves the radial equivalent; the literal normal and active increments give the summable physical bound. In general $W(r)$ is larger than the tracking error, so replacing this nonlinear center by the origin would lose the conclusion. The proof shrinks only the eventual neighborhood, never the admitted step size. ∎

A genuinely nonlinear example. At $x_* = (1/2, 1/2, 0, 0)$ put

$$u = x_3, \quad v = x_4, \quad r = (u + v)/2, \quad w = (u - v)/2, \quad y = (x_1 - x_2)/2,$$

and let

$$f = w^2 + y^2 + (r + w)^4 + (r + y)^4. \tag{M.50}$$

This globally convex polynomial has the unique simplex minimizer x_*. Its two full transverse centers solve $z + 2(r + z)^3 = 0$, giving

$$W(r) = E(r) = -2r^3 + O(r^5), \qquad f_0(r) = 2r^4 + O(r^6).$$

An admitted ambient bound is $L = 17$. For $\eta < 1/17$, the theorem yields $kr_k^4 \to 1/(16\eta)$ and $\|\Delta x_k\| = O(k^{-5/4})$. Supplement Section S4.6 verifies all these assertions.

The nonlinear center is larger than its tracking error. Static minimization alone would not justify replacing the exact projected dynamics by a reduced Burg map. The complete proof in Supplement Sections S4.1–S4.5 retains both actual source–target clocks. General two-normal qualitative classification and quartic exhaustion are not needed here and are preserved in the source archive.

8 Degenerate supported profiles and sharp endpoints

The following reduced-jet assumptions provide asymptotic information beyond the scalar length mechanism. The zero-tilt and tilted branches have different supported scales; their complete proofs retain the actual target weights and fixed ambient step.

8.1 The zero-tilt analytic branch with a positive normal kernel

A positive one-dimensional normal kernel can coexist with critical supported degeneracy. The appropriate transverse profile is the analytic minimizing center of the objective; it need not have the bias order of an exactly homogeneous model. The next theorem proves capture in the zero-tilt branch of a full reduced Taylor jet and keeps the original fixed Burg step. Section 8.2 classifies the remaining degree-$(m+1)$ tilt and proves its distinct active scale.

Physical chart and reduced jet. Let $A = \operatorname{supp} x_*$, $I = A^c$, $N = |I| \geq 2$, and $d = |A| - 1 \geq 1$. The columns of E span the entire supported-face tangent, with $E_I = 0$ and $\mathbf{1}^T E = 0$. Set

$$\begin{aligned} &S_* = \sum_{i \in A} x_{*,i}^2, \quad b_A = x_{*,A}^2 / S_*, \quad B_I = I_N, \quad B_A = -b_A \mathbf{1}_N^T, \\ &x = x_* + Bu + Ey, \quad u = x_I, \quad G = E_A^T \operatorname{diag}(x_{*,i}^{-2}) E_A \succ 0. \end{aligned} \tag{M.51}$$

Powers and reciprocals of coordinate vectors are componentwise. In particular $E_A^T \operatorname{diag}(x_{*,i}^{-2}) B_A = 0$. There is no omitted neutral supported block. For a fixed vector a with all $a_i > 0$, define

$$\ell_i = a_i^{-1}, \quad D_a = \operatorname{diag}(a_i), \quad r = \ell^T u / N, \quad P = I_N - a\ell^T / N, \quad w = Pu, \quad E_0 = \ker \ell^T. \tag{M.52}$$

Use fixed linear coordinates on E_0 when differentiating with respect to w; write the corresponding embedded vector when adding w to ra. No nonlinear change is made to the physical Burg geometry.

Write $F(w, r, y) = f(x) - f(x_*)$. Assume F is real analytic and convex on an open neighborhood of x_* in the affine simplex hyperplane, with zero affine gradient there. Its Hessian is positive definite on E_0 and zero on the entire (r, y) block, with no cross block.

Equivalently, the physical normal Hessian $K \succeq 0$ has kernel $\mathrm{span}\{a\}$ and the entire supported tangent is in the Hessian kernel. Let

$$F_w(\phi(r,y),r,y)=0, \qquad \phi(0)=D\phi(0)=0, \qquad f_{\mathrm{red}}(r,y)=F(\phi(r,y),r,y).$$

The analytic implicit-function theorem defines this center locally. For even $m \geq 4$, impose the *full mixed reduced jet*

$$f_{\mathrm{red}}(r,y)=cr^m+c_1r^{m+1}+\mathcal{J}_{m+2}(r,y)+O((|r|+\|y\|)^{m+3}), \tag{M.53}$$

where $c>0$, $\mathcal{J}_{m+2}$ is homogeneous, and $\mathcal{J}_{m+2}(0,y)>0$ for $y\neq 0$. All lower reduced jets vanish, the degree-m jet is exactly cr^m, and the degree-$(m+1)$ jet is exactly c_1r^{m+1}. Analytic differentiated remainders are understood. Separate radial and active axis orders do not suffice. Joint convexity of $\mathcal{J}_{m+2}$ is not required.

Theorem 9 (Automatic analytic capture and sharp physical asymptotics)**.** *Under* (M.51)–(M.53)*, consider the exact fixed-step map* (M.1)*, with $\eta L<1$. Each of the following entry assumptions suffices:*

1. *The orbit converges to x_*, and L is an admitted local ambient smoothness bound on its eventual source–target segments.*
2. *The initial point is sufficiently close to x_* and interior, and L is an admitted bound on a sufficiently small convex neighborhood. Such starts stay locally trapped and converge to x_*.*
3. *The objective extends convexly and smoothly to the entire simplex, x_* is its unique constrained minimizer, and L is an admitted global ambient bound. Every interior start then converges to x_*.*

Local hypotheses alone do not imply the third version. In each version put

$$q=(m+2)/2, \quad \lambda=mc/N, \quad \lambda_1=(m+1)c_1/N,$$
$$C_r=(\eta m\lambda)^{-1/m}, \quad z=G^{-1}E_A^T(-1/x_A+1/x_{*,A}).$$

Then, on the tail, $u_i \asymp r \asymp k^{-1/m}$, $w/r\to 0$, and

$$\|y\|+\|z\|=O(r), \qquad w=\phi(r,y)+O_\eta(r^{m+1}), \tag{M.54}$$

$$\phi(r,y)=br^q+O((|r|+\|y\|)^{q+1}), \qquad kF(x_k)\longrightarrow \frac{N}{\eta m^2}, \tag{M.55}$$

$$\Delta u=-\eta(\lambda r^{m+1}+\lambda_1 r^{m+2})a+O_\eta(\eta r^{m+3}), \tag{M.56}$$

$$\Delta r=-\eta(\lambda r^{m+1}+\lambda_1 r^{m+2})+O_\eta(\eta r^{m+3}), \tag{M.57}$$

$$r_k=C_rk^{-1/m}[1+O(k^{-1/m})]. \tag{M.58}$$

Here $b \in E_0$ is fixed. If $c_1 = 0$, the relative error in (M.58) *improves to $O(k^{-2/m})$. Every literal normal is eventually strictly decreasing, and the orbit has finite Euclidean physical length.*

The sequence (r_k, v_k), where $v = z/r$, has finite length, and $v_k \to v_*$ with

$$V(v) = \mathcal{J}_{m+2}(1, v) - \tfrac{\lambda}{2} v^T G v, \qquad \nabla V(v_*) = 0. \tag{M.59}$$

The critical set need not be discrete. With $d_* = Ba + Ev_*$, whose normal block is $a \neq 0$,

$$k^{1/m}(x_k - x_*) \longrightarrow C_r d_*, \qquad k^{1+1/m}\Delta x_k \longrightarrow -\frac{C_r}{m} d_*, \tag{M.60}$$

$$\mathcal{L}_k \sim \|d_*\| r_k, \qquad \|x_k - x_*\| \sim \|d_*\| r_k. \tag{M.61}$$

In particular $\mathcal{L}_k / \|x_k - x_*\| \to 1$. Constants may depend on the fixed step, the germ and the orbit.

Mechanism and proof dependencies. Supplement Section S5 gives the complete proof. Convexity forces the first center jet to be purely radial: $\phi = br^q + O(s^{q+1})$, $\phi_y = O(s^q)$, where $s = |r| + \|y\|$. The same lemma shows that eliminating the positive transverse block changes the critical raw-slice jet only by a pure radial term (Section S5.2). Local coercivity, exact weighted descent and convex mirror comparison establish the stated entry versions and the independent coarse clock

$$F = O(T^{-1}), \quad r = O(T^{-1/m}), \quad y = O(T^{-1/(m+2)}),$$
$$\delta := w - \phi = O(T^{-1/2}), \qquad T = 1 + \eta(k - k_0).$$

Pairing the exact normal reciprocal identity with a gives

$$Q = B^T(1/x) = 1/u - (b_A^T(1/x_A))\mathbf{1}_N, \qquad \Delta(a^T Q) = \eta F_r.$$

The center estimates bound its signed accumulation by $CT^{1/m}$. Positivity of all a_i/u_i yields every lower normal bound and $w/r \to 0$, before active capture.

A preliminary transverse estimate is essential: the coarse $O(\delta^2)$ active forcing cannot yet be absorbed at $\|y\| \asymp r$. For the *actual* source–target multiplier, with $\omega_i = x_i x_i^+$ and $C_{y,A} = G^{-1} E_A^T \operatorname{diag}(x_{*,i}^{-2})$,

$$\mu = \frac{(C_{y,A}\omega_A)^T F_y + \omega_I^T F_u}{\sum_i \omega_i}. \tag{M.62}$$

Using $C_{y,A} x_{*,A}^2 = 0$ and retaining target increments gives the pre-capture bound $|\mu| \leq C_\eta[T^{-1} + r^2(\|\delta\| + |f_{\mathrm{red},r}|)]$. The literal normal identity is

$$\Delta u_i = -\eta u_i^2 (F_{u_i} - \mu) + \frac{\eta^2 u_i^3 (F_{u_i} - \mu)^2}{1 + \eta u_i (F_{u_i} - \mu)}. \tag{M.63}$$

The coordinate factor u_i^3 cannot be replaced by a uniform discretization error. Projection, positivity of $A_0 = (PD_a^2 K)|_{E_0}$ in the D_a^{-2} metric, and the *actual* center increment yield

$$\|\delta^+\|_* \leq (1 - c_0 \eta T^{-2/m})\|\delta\|_* + C_\eta \eta [T^{-1-2/m} + T^{-3/2+1/(m+2)}].$$

Thus $\delta = O(T^{-\beta})$, where $\beta = 5/6$ if $m = 4$ and $\beta = 1$ if even $m \geq 6$. This makes $F_y - f_{\mathrm{red},y} = o(T^{-1-1/m})$.

The exact dual step $\Delta z = -\eta G^{-1} F_y$, active positivity of the full critical jet, and its analytic inverse chart now imply

$$R^+ \leq R - c_0 \eta R^{m+1} + C_\eta \eta T^{-1-1/m}, \qquad R = \|z\|_G.$$

A scalar comparison proves $y, z = O(r)$. Repeating the transverse contraction at this scale, integrating $D\phi$ along each actual source–target segment, proves the final lag $\delta = O_\eta(r^{m+1})$. Substitution in (M.63) gives the two radial coefficients and the stated error; summation of the power-reciprocal increment gives (M.58). The different errors for $c_1 \neq 0$ and $c_1 = 0$ come from summing r and r^2, respectively. These arguments occupy Sections S5.4–S5.9.

With $\alpha = \eta r^m$, the bounded profile obeys

$$\Delta v = -\alpha G^{-1} \nabla V(v) + O_\eta(\alpha r), \qquad \Delta r = -\lambda \alpha r + O_\eta(\alpha r^2). \tag{M.64}$$

A sufficiently large fixed lift $V(v) + L_1 r^2$ has full-tail descent and a source step bound. Compact-cluster uniformization of the ordinary analytic gradient inequality gives finite length and point selection in (r, v) (Section S5.10); no unproved retention assumption or isolated-critical-point hypothesis is used. This imports ordinary analytic KL [2–4]. The actual inverse dual chart then gives the physical increment independently of the position equivalent. Positive radial decrements telescope to prove the tail equivalent (Section S5.11). ∎

The exact homogeneous class as a stronger-bias corollary. On the entire feasible simplex suppose exactly

$$F(y, u) = \tfrac{1}{2} u^T K u + G_m(u) + J_{m+2}(y, u), \tag{M.65}$$

where G_m and J_{m+2} are globally convex homogeneous polynomials of the indicated even degrees, $G_m(a) > 0$, and $J_{m+2}(y, 0) > 0$ for $y \neq 0$. There is no analytic remainder in this subcase. Define

$$A_0 = (P D_a^2 K)|_{E_0}, \qquad b_0 = P D_a^2 \nabla G_m(a), \qquad w_* = -A_0^{-1} b_0, \qquad \lambda = m G_m(a)/N.$$

Corollary 8.1 (Exact homogeneous transverse bias)**.** *Every interior-start orbit under* (M.65) *and the original ambient $\eta L < 1$ has all conclusions of Theorem 9. More precisely,*

$$r_k = C_r k^{-1/m}[1 + O(k^{-2/m})], \qquad u_k = r_k a + r_k^{m-1}[w_* + O(r_k^2)]. \tag{M.66}$$

Here $c = G_m(a)$, $c_1 = 0$, and the angular potential can be written $J_{m+2}(v, a) - \lambda v^T G v/2$. Also

$$\Delta u = -\eta \lambda r^{m+1} a + O_\eta(\eta r^{m+3}), \qquad \Delta r = -\eta \lambda r^{m+1} + O_\eta(\eta r^{m+3}). \tag{M.67}$$

Proof. All objective terms are nonnegative, and their common zero is x_*, so global entry applies. Stationarity of the transverse center gives $\phi(r,y) = r^{m-1}w_* + O((|r| + \|y\|)^{m+1})$. Its reduced critical jet is

$$\mathcal{J}_{m+2}(r,y) = J_{m+2}(y, ra) - \mathbf{1}_{\{m=4\}}\tfrac{1}{2}(w_*^T K w_*)r^6.$$

The quartic Schur correction is purely radial and hence changes the angular potential only by a constant. This calculation, including the missing degree-$(m+1)$ jet, is proved in Supplement Section S5.12. Combining the center expansion with the theorem's lag gives the stronger bias. That section also retains the explicit weighted normal expansion and quotient tracking of this subcase. ∎

For $N = 2$, take $a = (1,1)^T$, $K = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, $\gamma_1, \gamma_2, \sigma > 0$, and any $d \times 2$ matrix T. The choice

$$G_m(u) = \frac{\gamma_1 u_1^m + \gamma_2 u_2^m}{m}, \qquad J_{m+2}(y,u) = \frac{[(y-Tu)^T G(y-Tu) + \sigma^2\|u\|^2]^{(m+2)/2}}{m+2}$$

is globally convex. It has $\lambda = (\gamma_1 + \gamma_2)/2$ and $w_* = (\gamma_2 - \gamma_1)(1,-1)^T/4$. If $w_* = 0$, the stronger statement gives $w = O(r^{m+1})$.

Sharpness and scope. The full mixed-jet condition is restrictive; it is not a genericity assertion for convex analytic germs. A convex analytic center can begin at $r^{(m+2)/2}$, earlier than the homogeneous r^{m-1} bias. An explicit entire globally convex polynomial with the same pure-axis leading orders but a degree-$(m+1)$ term $r^m y/4$ has invariant-slice orbits with $y \neq O(r)$. Symmetric convex analytic perspectives also show that allowed critical mixed jets change the angular critical set. Full calculations are in Supplement Section S5.13.

The present theorem admits analytic remainders subject to (M.53), but excludes higher-dimensional normal kernels, a kernel ray with zero components, an omitted neutral block, and arbitrary lower-order active-dependent jets. Open-neighborhood convexity supplies the center lemma and local dynamic comparisons; its static feasible-tube extension alone is not a feasible-only dynamic theorem. Standard convex mirror convergence, ordinary KL, and the classical gradient-geometric tail ratio are imported precedents [1, 2, 7].

8.2 The convexity-forced tilted branch

The degree-$(m+1)$ restriction in (M.53) has an exhaustive alternative. Retain the physical chart, positive normal kernel, analyticity and open-neighborhood convexity of Section 8.1, but assume only

$$\begin{gathered} f_{\mathrm{red}}(r,y) = cr^m + \mathcal{P}_{m+1}(r,y) + \mathcal{J}_{m+2}(r,y) + O(s^{m+3}), \\ H(y) := \mathcal{J}_{m+2}(0,y) > 0 \quad (y \neq 0), \end{gathered} \tag{M.68}$$

where $s = |r| + \|y\|$, $c > 0$, and the entire degree-m jet is radial. Convexity forces

$$\mathcal{P}_{m+1}(r, y) = c_1 r^{m+1} + r^m b \cdot y. \tag{M.69}$$

Indeed, the leading active Hessian of $f_{\mathrm{red}}(\varepsilon r, \varepsilon y)$ is $\varepsilon^{m-1}(\mathcal{P}_{m+1})_{yy}$. Its positivity for both signs of ε forces this polynomial to vanish. Homogeneity gives (M.69). The vector $b \in \mathbb{R}^d$ here is the active tilt, distinct from the radial transverse-center coefficient in (M.55).

The branch $b = 0$ is exactly Theorem 9, with its sharper $y = O(r)$, angular potential, finite lifted length, two-term normal expansion and physical tail ratio. For $b \neq 0$, set

$$\begin{gathered} p = m + 2, \quad q = p/2, \quad \lambda = mc/N, \quad \theta = \frac{m}{m+1}, \quad \rho = r^\theta, \quad t = r^{1/(m+1)}, \\ \beta = \min\{1, \tfrac{3}{2} - \tfrac{1}{m+2} - \tfrac{2}{m}\}, \qquad \kappa = \min\{\theta, m\beta - m + 1\} > 0. \end{gathered} \tag{M.70}$$

Thus $(\beta, \kappa) = (5/6, 1/3)$ for $m = 4$, and $(1, \theta)$ for even $m \geq 6$.

Theorem 10 (Tilted analytic capture and unique forced profile)**.** *Assume* (M.68) *with* $b \neq 0$ *and any one of the three entry versions of Theorem 9, retaining the original ambient* $\eta L < 1$*. With* $T = 1 + \eta(k - k_0)$ *and* $\delta = w - \phi(r, y)$*, every admitted orbit satisfies*

$$u_i \asymp r \asymp T^{-1/m}, \qquad \delta = O_\eta(T^{-\beta}), \quad y = O(\rho), \quad w = O(r^q), \tag{M.71}$$

$$\Delta u = -\eta\lambda r^{m+1} a + O_\eta(\eta r^{m+1+\kappa}), \tag{M.72}$$

$$\Delta r = -\eta\lambda r^{m+1} + O_\eta(\eta r^{m+1+\kappa}), \tag{M.73}$$

$$r_k = (\eta m \lambda k)^{-1/m}[1 + O(k^{-\kappa/m})], \quad kF(x_k) \longrightarrow \frac{N}{\eta m^2}. \tag{M.74}$$

Every literal normal is eventually strictly decreasing. The polynomial H *is convex and strictly convex, and* $H(v) + b \cdot v$ *has a unique minimizer* $v_b \neq 0$*. For the exact supported dual coordinate* z *of Theorem 9,*

$$\frac{z_k}{\rho_k}, \ \frac{y_k}{\rho_k} \longrightarrow v_b, \qquad \nabla H(v_b) + b = 0, \qquad x_k - x_* = \rho_k E v_b + o(\rho_k). \tag{M.75}$$

The physical sequence has finite Euclidean length, including when $\nabla^2 H(v_b)$ *is singular. In particular* $\|x_k - x_*\| \sim \|E v_b\| \rho_k$*.*

Mechanism and complete proof location. Sections S5.14–S5.18 of the supplement give the full extension. The static-center lemma already applies to $f_{\mathrm{red}} = cr^m + O(s^{m+1})$. The exact reciprocal lower bound and preliminary transverse contraction are rechecked with the tilt before any active capture is used. The active forcing $r^m b$ changes the scalar capture inequality to

$$R^+ \leq R - c_0 \eta R^{m+1} + C_\eta \eta T^{-1}, \qquad R = \|z\|_G,$$

which yields $y, z = O(T^{-1/(m+1)}) = O(\rho)$. The literal rational normal step then proves (M.72)–(M.74).

Retain the complete analytic moving potential

$$\Psi(t,v)=\frac{f_{\mathrm{red}}(t^{m+1},t^m v)-ct^{m(m+1)}}{t^{m(m+2)}},\qquad \Psi(0,v)=H(v)+b\cdot v. \tag{M.76}$$

For $v=z/\rho$ and $h_k=\eta r_k^{m-\theta}$, the exact map gives

$$\Delta v=-hG^{-1}\nabla_v\Psi(t,v)+O_\eta(h\rho),\qquad h\to 0,\quad \sum h=\infty.$$

Strict convexity gives attraction to v_b by a compact-annulus argument. Since t is eventually decreasing and $\sum h\rho^2<\infty$, the moving-potential descent estimate gives $\sum h\|\nabla_v\Psi\|^2<\infty$. Cauchy–Schwarz then gives $\sum\rho\|\Delta v\|<\infty$, and the inverse dual chart proves physical finite length. Replacing Ψ by its limiting potential would discard a generally nonsummable moving term; unweighted finite length of v is not asserted. ∎

Theorem 11 (Sharp endpoints with a positive forced Hessian)**.** *Under Theorem 10, assume $A_b:=\nabla^2H(v_b)\succ 0$. There is an analytic $V(t)$ with $V(0)=v_b$ such that $\bar y(r)=\rho V(t)$ and $f_{\mathrm{red},y}(r,\bar y(r))=0$. Then*

$$y-\bar y(r)=\rho^2[\theta\lambda A_b^{-1}Gv_b+o(1)], \tag{M.77}$$

$$\Delta x=-\eta\theta\lambda r^{m+\theta}Ev_b+o(\eta r^{m+\theta}), \tag{M.78}$$

$$\mathcal{L}_k\sim\|Ev_b\|\rho_k\sim\|x_k-x_*\|. \tag{M.79}$$

For one active coordinate and nonzero tilt, the positive-Hessian hypothesis is automatic.

Proof. Supplement Section S5.19 compares the exact dual orbit with the dual image of $\bar y(r)$ at the current full normal coordinate. Its stable lag recurrence proves (M.77), then the literal force proves (M.78). Finally $\rho-\rho^+=\eta\theta\lambda r^{m+\theta}[1+o(1)]$ telescopes to (M.79). No position equivalent is differentiated. ∎

Theorem 12 (Sharp endpoints with one forced-Hessian kernel direction)**.** *Under Theorem 10, suppose* $\dim\ker\nabla^2H(v_b)=1$. *For the physical scaled coordinate $Y=y/\rho$,*

$$\Delta Y=o(\eta r^m),\qquad k\Delta Y\longrightarrow 0. \tag{M.80}$$

The literal physical-step and sharp tail equivalents (M.78)–(M.79) *follow. The stable active block may have arbitrary dimension. The theorem does not assert the full-vector static-lag formula* (M.77)*.*

Force cancellation and proof location. Supplement Section S5.21 proves the complete result. In physical scaled coordinates the exact recurrence improves to

$$\Delta Y=-hG^{-1}\nabla\Psi(t,Y)+\theta\lambda h\rho Y+o(h\rho),\qquad h=\eta r^{m-\theta}.$$

A fixed metric whitening and analytic elimination of the stable active block leave a scalar increasing force. Its unique nearby force-level roots have convergent Puiseux expansions, so their motion is $o(h\rho)$. Exact moving-barrier entry and invariance prove cancellation of that scalar force. Stable tracking then proves $\Delta Y=o(h\rho)$. Reconstructing the literal physical step and telescoping positive ρ decrements gives the two endpoints separately. ∎

Admissibility, invariance and scope. The raw slice and transverse reduction have the same c, c_1, b, H: their first difference is a pure radial multiple of r^{m+2}. Necessary radial Schur compensation is

$$(\mathcal{J}_{m+2})_{yy}(1,0) \succeq \frac{m}{c(m-1)} bb^T.$$

This is a necessary condition, not a sufficient coefficientwise test. The class is nonempty: for $d_0 = b/[c(m-1)]$, the local convex analytic perspective

$$\tfrac{1}{2} w^T K_w w + \frac{c r^m}{(1 - d_0 \cdot y)^{m-1}} + H(y), \qquad K_w \succ 0,$$

has the prescribed tilt wherever the denominator is positive. Supplement Section S5.14 gives the precise local density and weighted invariance statements. The entire polynomial obstruction in Section S5.13 is now positively classified in Section S5.20.

The class still requires the full radial degree-m reduced jet, a strictly positive one-dimensional normal kernel, the coercive degree-$(m+2)$ pure-active jet, the complete supported tangent, and open-neighborhood convexity. It does not cover arbitrary lower mixed jets, higher-dimensional normal kernels, unrelated active degrees or merely feasible-side convexity. The reflection-symmetric mixed-rate criterion in Section 9 excludes the present nonzero tilt. Strict convexity gives uniqueness but does not imply $A_b \succ 0$; for example $H(v) = \sum v_j^{m+2}$ with $b = (-1, 0, \ldots)$ has a singular forced Hessian.

The sharp speed and tail ratio are proved here when the forced Hessian has corank zero or one. They remain open for corank at least two in this class; unique profile and finite physical length already hold there. The scalar Puiseux input is classical [11, Proposition 1.8.4]. Standard KL descent or continuous gradient secant results alone do not imply the literal discrete endpoints [1, 10]; the supplement gives a bounded applicability comparison.

9 Mixed rate classification and boundaries

A fixed active-face polynomial and a fixed leading normal term do not determine the active rate type. The following finite criterion separates algebraic, logarithmically corrected and stretched-exponential relaxation. Convexity must be checked independently of coefficient positivity.

9.1 The class and the finite rate criterion

Write the three-coordinate simplex in the affine coordinates

$$\begin{gathered} x_1 = \frac{1-u}{2} + y, \qquad x_2 = \frac{1-u}{2} - y, \qquad x_3 = u, \\ D = \left\{ (y,u) : 0 \le u \le 1,\ |y| \le \frac{1-u}{2} \right\}. \end{gathered} \tag{M.81}$$

The variable y is an active-face displacement; u is the literal normal coordinate. Let

$$F(y,u)=\sum_{a,b\ge 0} c_{ab}y^{2a}u^{b}, \qquad c_{ab}\ge 0, \quad c_{00}=0, \tag{M.82}$$

be a finite polynomial, with integer indices. Assume that F is convex on D and that

$$\begin{aligned} F(y,0)&=a_p y^p+O(y^{p+2}),\\ F(0,u)&=a_m u^m+O(u^{m+1}),\end{aligned} \qquad p\ge 4 \text{ even}, \quad m\ge 1, \quad a_p,a_m>0. \tag{M.83}$$

Convexity is a separate hypothesis on the whole polynomial. Neither coefficient positivity nor the support inequality proved below is a substitute for it. In particular, the main theorem permits odd m and odd normal powers whenever convexity on D is verified. For example, $y^4+u^3/3$ is admissible on D, though it is not globally convex on $\mathbb{R}^2$. The case $m=1$ is included without a limiting convention.

Use the original exact Burg iteration. For an admitted ambient Euclidean gradient-Lipschitz bound $L>0$ on the simplex, retain precisely

$$\frac{1}{X_i}-\frac{1}{x_i}=\eta\big(g_i(x)-\mu\big), \qquad \sum_i X_i=1, \qquad 0<\eta L<1, \tag{M.84}$$

where $x=x_k$, $X=x_{k+1}$, and g is the objective gradient. The affine extension $f(x)=F((x_1-x_2)/2,x_3)$ is one representative. Representatives agreeing on the simplex differ in their gradients there only by a common-coordinate scalar and give the same map. No new step bound in (y,u) coordinates will be imposed.

Theorem 13 (Normal profile and complete active classification)**.** *Under* (M.82)–(M.84)*, every interior start converges to* $x_*=(1/2,1/2,0)$ *and has finite physical Euclidean length. Put*

$$\beta=\frac{1}{m}, \qquad C=(\eta m^2 a_m)^{-1/m}. \tag{M.85}$$

For some $\delta>0$ *depending on the polynomial, with implicit constants allowed to depend on the orbit and step,*

$$\begin{aligned} u_k&=Ck^{-1/m}\big[1+O(k^{-\delta})\big],\\ k^{1+1/m}(u_{k+1}-u_k)&\longrightarrow -\frac{C}{m}, \qquad kF(y_k,u_k)\longrightarrow\frac{1}{\eta m^2}.\end{aligned} \tag{M.86}$$

The normal coordinate is eventually strictly decreasing. The active sign is preserved from the first step. If $y_0=0$*, then* $y_k=0$ *for all* k*.*

For $y_0\neq 0$*, set* $\sigma=\operatorname{sgn}(y_0)$ *and define, for each supported active monomial,*

$$s_{ab}=2a-2, \qquad A_{ab}=\frac{\eta}{8}(2a)c_{ab}C^b \quad (a\ge 1), \qquad \nu=\min\{b: c_{1b}>0\}, \tag{M.87}$$

with $\nu=+\infty$ *if the set is empty. When finite,* $\nu\ge 1$*.*

If $\nu < m$, then

$$\frac{-\log|y_k|}{k^{1-\nu/m}} \longrightarrow \frac{A_{1\nu}}{1-\nu/m}. \tag{M.88}$$

This is a logarithmic equivalent for a stretched exponential, not a claim of a constant multiplicative prefactor after removal of its leading clock.

If $\nu \geq m$, put $\kappa = A_{1m}$ when that monomial is present, and $\kappa = 0$ otherwise. Define

$$\begin{aligned} \alpha &= \max_{\substack{a\geq 2,\ b<m \\ c_{ab}>0}} \frac{1-b/m}{2a-2}, \\ I &= \{(a,b) : a \geq 2,\ c_{ab} > 0,\ b/m + \alpha(2a-2) = 1\}, \\ P(v) &= \sum_{(a,b)\in I} A_{ab} v^{2a-2} \quad (v \geq 0). \end{aligned} \tag{M.89}$$

The maximum set and I are nonempty. The alternatives are:

1. *If $\kappa < \alpha$, there is one $V > 0$ satisfying $P(V) = \alpha - \kappa$, and*

$$y_k \sim \sigma V k^{-\alpha}. \tag{M.90}$$

2. *If $\kappa > \alpha$, there is an orbit-dependent $D_0 \in (0, \infty)$ such that*

$$y_k \sim \sigma D_0 k^{-\kappa}. \tag{M.91}$$

3. *If $\kappa = \alpha$, let*

$$s = \min_{(a,b)\in I}(2a-2), \qquad A_* = \sum_{\substack{(a,b)\in I \\ 2a-2=s}} A_{ab}. \tag{M.92}$$

Then

$$y_k \sim \sigma k^{-\alpha}[sA_* \log k]^{-1/s}. \tag{M.93}$$

The pure active term makes the maximum in (M.89) nonempty and $\alpha > 0$. The harmonic coefficient has the useful expression

$$\kappa = \frac{c_{1m}}{4m^2 a_m}, \tag{M.94}$$

with $c_{1m} = 0$ when absent. Thus the comparison of κ and α is independent of the admitted fixed step, whereas amplitudes and stretched-exponential clocks can depend on it.

The exact dual chart and its monotone coordinates in Supplement Lemma S6.1 give finite length and rational normalization independently of the rate criterion. The criterion then determines the exact normal order and coefficient, the power remainder and literal speed, and the active rate type despite mixed forces. Its complete proof is in Supplement Section S6.

9.2 Small convex perturbations can change the active rate type

Proposition 9.1 (Compact-domain perturbation instability)**.** Let $d \geq 2$ be an integer and q, m be positive even integers satisfying $q(d-1) < m < qd$. For $\varepsilon \geq 0$, set

$$F_\varepsilon(y,u) = \frac{(y^2+\varepsilon u^q)^d}{2d} + \frac{u^m}{m}. \tag{M.95}$$

Every member is globally convex. As $\varepsilon \downarrow 0$, $F_\varepsilon \to F_0$ in every fixed C^r norm on the compact simplex. A common positive step satisfying the original condition can be chosen for all sufficiently small ε.

For every member, the entire active-face restriction is $y^{2d}/(2d)$, the leading normal-axis term is u^m/m, and the leading normal and objective constants are unchanged. For every nonzero active start, the active law at $\varepsilon = 0$ is

$$y_k \sim \operatorname{sgn}(y_0)\left[\frac{\eta(2d-2)}{8}\,k\right]^{-1/(2d-2)}. \tag{M.96}$$

For every fixed $\varepsilon > 0$, it instead satisfies, with $C = (m\eta)^{-1/m}$ and $\nu = q(d-1)$,

$$\frac{-\log|y_k|}{k^{1-\nu/m}} \longrightarrow \frac{\eta\varepsilon^{d-1}C^\nu}{8(1-\nu/m)}. \tag{M.97}$$

The full normal-axis polynomial is not preserved:

$$F_\varepsilon(0,u) = \frac{u^m}{m} + \frac{\varepsilon^d}{2d}u^{qd}. \tag{M.98}$$

Proof. The convex composition proof of Proposition S6.6 works for every $\varepsilon \geq 0$. The finite binomial expansion has coefficients converging to those of F_0; hence all derivatives through any fixed integer order r converge uniformly on D. The ambient Hessians of the affine extensions are uniformly bounded on the compact simplex for $0 \leq \varepsilon \leq \varepsilon_0$. Taking one positive upper bound L for these norms supplies a common η with $0 < \eta L < 1$.

At $\varepsilon = 0$, the only active derivative term is y^{2d-1}, giving (M.96). For $\varepsilon > 0$,

$$(F_\varepsilon)_y = y(y^2+\varepsilon u^q)^{d-1}$$

has smallest quadratic stiffness power ν and coefficient ε^{d-1}. Since $\nu < m$, Theorem 13 gives (M.97). Direct restriction gives the face polynomial and (M.98); its extra term is higher order because $qd > m$. ∎

Thus global convexity of each perturbation is compatible with arbitrarily small $C^r(D)$ changes and a change of active rate type. This is not smallness in a norm on all of $\mathbb{R}^2$, preservation of both full axis polynomials, or a claim of a crossover time uniform as $\varepsilon \downarrow 0$. The normal universality here concerns the fixed pure-normal leading data within the separately verified convex, positive-coefficient class; an arbitrary added positive mixed monomial need not preserve convexity.

10 Boundaries of the conclusions

The general convex theorem allows arbitrary normal counts and KKT gaps, but retains unique constrained minimality, normal growth and a positive supported Hessian for the full physical lift. Positive-power normal variation survives supported degeneracy. What remains unresolved in general is the actual center variation or another supported-motion estimate when that center is multivariate and non-Lipschitz. The stationary and one-signed moving classes above close this issue only under their stated structural identities and restoration bounds.

The planar analytic theorem has a different scope: it selects a KKT point without convexity, but only on a closed triangle and with analyticity across its full boundary. Its proof does not imply arbitrary-dimensional nonconvex selection. Nonunique minimizing sets with possible support loss and unrestricted degenerate supported faces are not covered by the general theorem. No universal limiting direction or sharp tail/distance equivalent follows from finite length alone.

The finer results separate these issues. Oriented flags control normal comparability; the split-clock example shows that comparability is not required for rectifiability. The nonlinear-center theorem derives a radial equivalent under a rank-one normal cancellation hypothesis. The tilted theorems require a specified full reduced jet and positive or corank-one forced Hessian for their endpoint equivalents. The mixed polynomial criterion concerns its reflection-symmetric convex class and permits rate-type changes under small convex perturbations. Higher-corank sharp endpoints and general degenerate scalar-center variation remain separate questions.

10.1 A degenerate face need not inherit the normal tail exponent

Consider the globally convex polynomial

$$f(x) = (x_1 - x_2)^4 + x_3$$

on the three-coordinate simplex. Its unique minimizer is $(1/2, 1/2, 0)$, the supported Hessian vanishes, and normal growth holds with $\beta = 1$ because $u = x_3 \le F$. Write $t = x_1 - x_2$ and

$$q = \frac{1}{x_2} - \frac{1}{x_1} = \frac{4t}{(1-u)^2 - t^2}.$$

The exact reciprocal update gives

$$q_{k+1} = q_k - 8\eta t_k^3. \tag{M.99}$$

Choose u_0 and $q_0 > 0$ sufficiently small. For instance $u_0 < 1/10$, $q_0 < 1/2$, and $0 < \eta < 1/24$ suffice. As long as these bounds hold, $t \le q/4 < 1/8$, all supported gradients are less

than 1 in absolute value, and therefore the weighted-average multiplier satisfies $\mu < 1$. Thus u decreases. Moreover

$$\frac{8\eta t^3}{q} = 2\eta t^2((1-u)^2 - t^2) < 1,$$

so q remains positive and decreases. These estimates close the invariant small region. Since the supported weights remain uniformly positive, $1-\mu$ is bounded below by a positive constant there; hence $u \to 0$. If q had a positive limit, then t would also have a positive limit and (M.99) would give a nonvanishing negative increment, a contradiction. Therefore $q, t \to 0$, $\mu \to 0$, and the normal reciprocal equation yields $u_k \sim 1/(\eta k)$.

Also $t_k/q_k \to 1/4$, so (M.99) becomes

$$q_{k+1} = q_k - (\eta/8 + o(1))q_k^3.$$

It follows that $q_k^{-2}/k \to \eta/4$ and

$$t_k \sim \frac{1}{2\sqrt{\eta k}}.$$

Consequently the normal tail is $\mathrm{TV}_N(n) = u_n = O(n^{-1})$, whereas

$$\mathcal{L}_n \geq \|x_n - x_*\|_2 \geq \frac{|t_n|}{\sqrt{2}} \asymp n^{-1/2}.$$

Thus the full physical tail cannot satisfy $O(n^{-1})$ in this degenerate-supported example. The lower bound is valid whether the tail length is finite or infinite, and directly checks the scope of Theorem 1.

Proof organization. The complete arbitrary-codimension proof, constrained analytic growth, positive-power variation and exact scalar lift are in the main text. The supplement contains the planar analytic division proof, stationary and moving cancellation estimates, flag and split-clock certificates, nonlinear two-clock tracking, tilted endpoints and mixed-rate calculations. Every active theorem has a complete proof in this pair. A noncompiled archive preserves the preceding source, including older qualitative alternatives and profile families omitted from the current reader exposition.

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